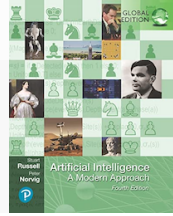
Fundamentals of AI and KR - Module 3

3. Building Bayesian networks

Roberta Calegari

Fall 2026

Constructing Bayesian networks



Constructing Bayesian networks I

Constructing a Bayesian network from an ordering

Assume that the variables are ordered as

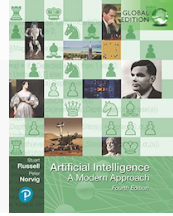
$$X_1, \ldots, X_n.$$

For each variable X_i, choose a set of parents $\mathrm{Parents}(X_i) \subseteq \{X_1, \ldots, X_{i-1}\}$ such that all other preceding variables are conditionally irrelevant once the parents are known:

$$X_i \perp \left(\{X_1, \ldots, X_{i-1}\} \setminus \mathrm{Parents}(X_i)\right) \mid \mathrm{Parents}(X_i).$$

Equivalently:

$$P(X_i \mid X_1, \ldots, X_{i-1}) = P(X_i \mid \mathrm{Parents}(X_i)).$$



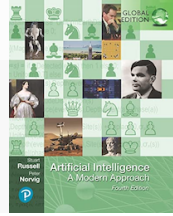
Constructing Bayesian networks II

Why this construction works

By the chain rule:

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid X_1, \ldots, X_{i-1})$$
$$= \prod_{i=1}^{n} P(X_i \mid \text{Parents}(X_i)).$$

The resulting network therefore represents the full joint probability distribution.



Constructing Bayesian networks III

What this construction does, and does not do

The procedure **guarantees** that the
resulting network factorizes the full joint probability distribution
correctly.

However, it
does not tell us how to choose a useful and compact set of parents.
In practice, parent selection relies on domain knowledge (preferably about causal mechanisms) or is learned from data.

Example



Let's construct a Bayesian network that helps us win a prize

Monty Hall puzzle

We're guests on a TV game show. We stand in front of three closed doors. A prize hides behind one of them. We choose the door on the left. At this point, the host, who knows where prize is, opens the middle door, to reveal it is empty. We are offered to modify our choice. *Should we?*

A Bayesian-network model of Monty Hall I



To answer this question, we model three random variables:

- $Prize \in \{left, middle, right\}$: the door hiding the prize;
- $Choice \in \{left, middle, right\}$: the contestant's initial choice;
- $HostOpens \in \{left, middle, right\}$: the door opened by the host.

Crucial domain knowledge

The **host knows the prize location** and never opens either the chosen door or the door hiding the prize. If two doors are possible, the host chooses between them uniformly at random.

A Bayesian-network model of Monty Hall II



Key modelling insight

The host's action depends on both *Prize* and *Choice*:

$$Prize \longrightarrow HostOpens \longleftarrow Choice.$$

This dependence is precisely what makes the observed open door informative.

The Monty Hall network

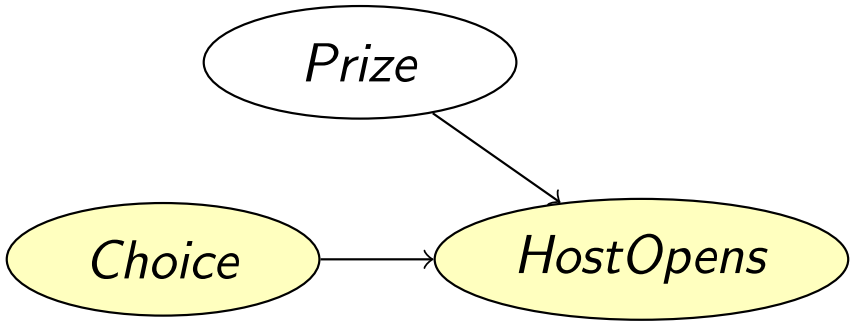


Observed situation

The contestant chooses the left door. The host opens the middle door and reveals that it is empty.

The host knows where the prize is. Therefore, the door opened by the host depends on:

- the location of the prize;
- the contestant's initial choice.



Observed evidence:

$$Choice = left, \qquad HostOpens = middle.$$

Key idea

The observed action of the host provides information about the unobserved location of the prize.

Updating the belief about the prize I



Observed evidence

The contestant chose the left door and the host opened the middle door:

$$Choice = left, \qquad HostOpens = middle.$$

How likely is this evidence?

$$\begin{aligned} Prize = left &\Rightarrow P(HostOpens = middle) = \tfrac{1}{2} \\ Prize = middle &\Rightarrow P(HostOpens = middle) = 0 \\ Prize = right &\Rightarrow P(HostOpens = middle) = 1 \end{aligned}$$

If the prize is behind the left door, the host can open either remaining door. If it is behind the right door, the host must open the middle door.

Updating the belief about the prize II



Combine prior and evidence

Each prize location has prior probability 1/3.

$$\begin{aligned} \textit{Prize} = \textit{left} &: \quad \tfrac{1}{3} \cdot \tfrac{1}{2} = \tfrac{1}{6} \\ \textit{Prize} = \textit{middle} &: \quad \tfrac{1}{3} \cdot 0 = 0 \\ \textit{Prize} = \textit{right} &: \quad \tfrac{1}{3} \cdot 1 = \tfrac{1}{3} \end{aligned}$$

These are **unnormalized** posterior values.

Updating the belief about the prize III



Normalize

$$\frac{1}{6} + 0 + \frac{1}{3} = \frac{1}{2}.$$

$$P(Prize = left \mid evidence) = \frac{1}{3}, \qquad P(Prize = right \mid evidence) = \frac{2}{3}.$$

What the evidence tells us

After updating the prior beliefs:

$$P(Prize = left \mid evidence) = \frac{1}{3}, \qquad P(Prize = right \mid evidence) = \frac{2}{3}.$$

The contestant should therefore switch to the right door.

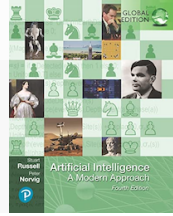
Updating the belief about the prize IV



Key idea

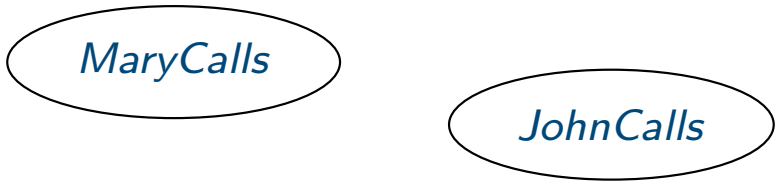
An **observation updates an agent's beliefs** about **unobserved causes** when the observation depends on those causes.

Here, the door opened by the host depends on the prize location. Observing the middle door open therefore changes the belief about where the prize is.

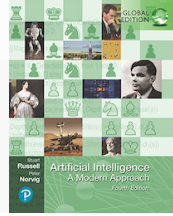


Example

Suppose we choose the ordering M, J, A, B, E

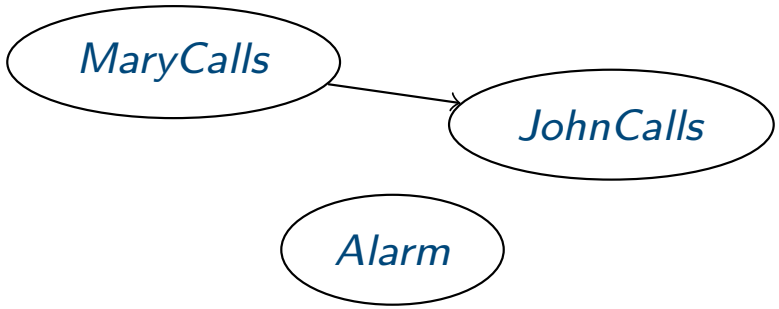


- $P(J|M) = P(J)$?

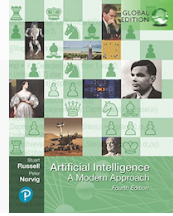


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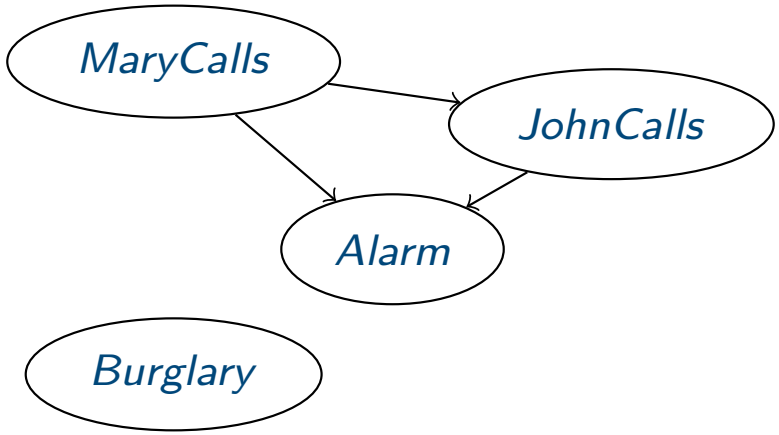


- $P(J|M) = P(J)$? No
- $P(A|J, M) = P(A|J)$? $P(A|J, M) = P(A)$?

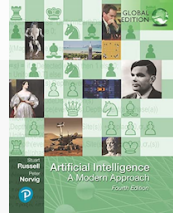


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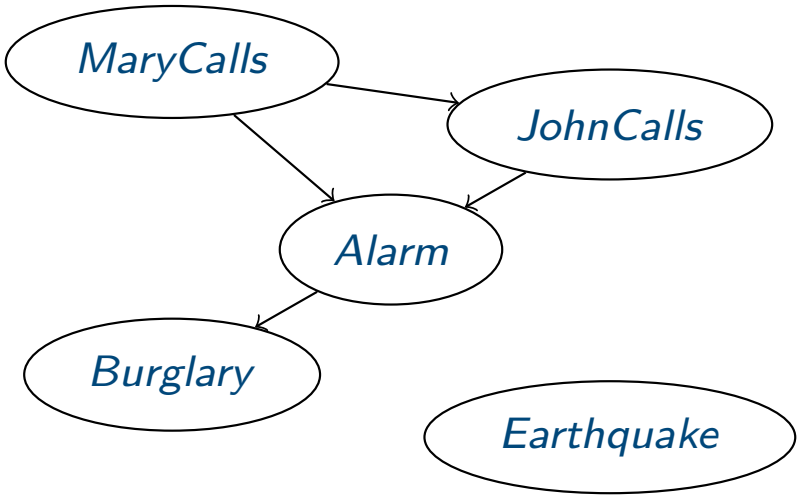


- $P(J|M) = P(J)$? No
- $P(A|J, M) = P(A|J)$? $P(A|J, M) = P(A)$? No
- $P(B|A, J, M) = P(B|A)$?
- $P(B|A, J, M) = P(B)$?

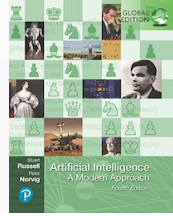


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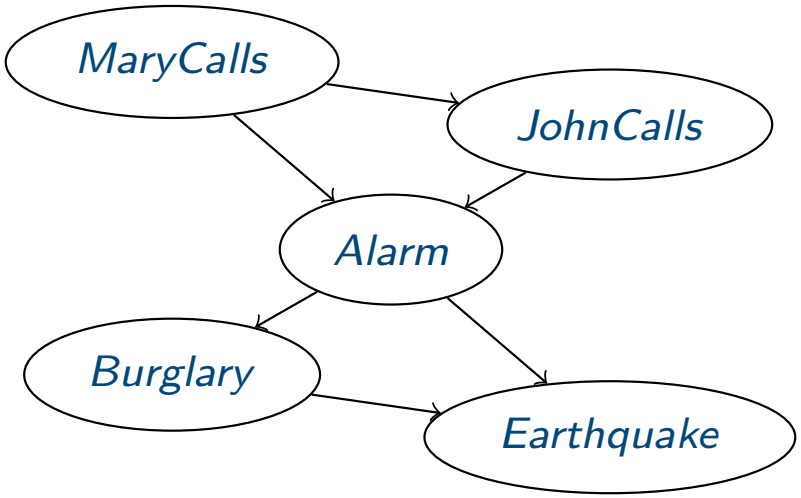


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- $P(B|A, J, M) = P(B|A)$? Yes
- $P(B|A, J, M) = P(B)$? No
- $P(E|B, A, J, M) = P(E|A)$?
- $P(E|B, A, J, M) = P(E|A, B)$?

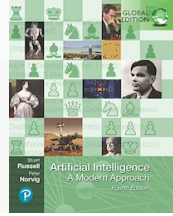


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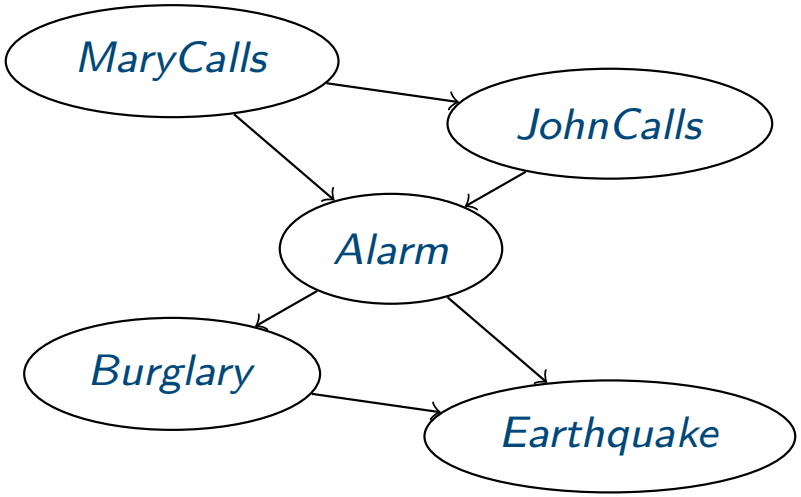


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- $P(E|B, A, J, M) = P(E|A, B)$? Yes



Example

Suppose we choose the ordering M, J, A, B, E

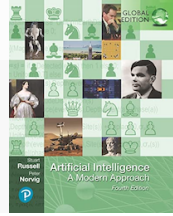


Why a causal ordering is often preferable

Variables ordered from possible causes to effects, usually makes it easier to:

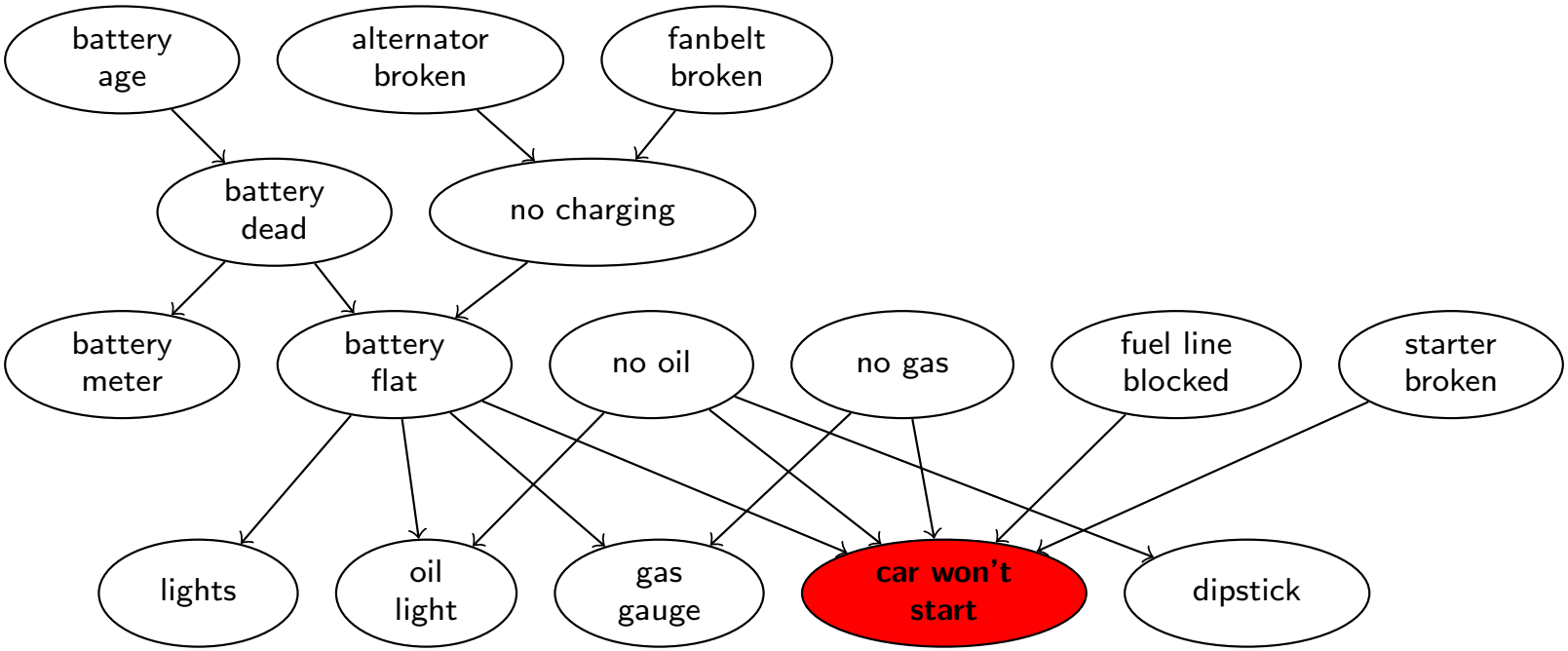
- identify which variables can be treated as conditionally independent;
- assess the required conditional probabilities;
- obtain a compact network with few parents per node.

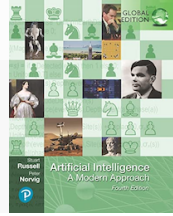
For the burglary example, the causal ordering requires 10 free parameters, whereas the noncausal ordering shown here requires 13.



Example: Car diagnosis

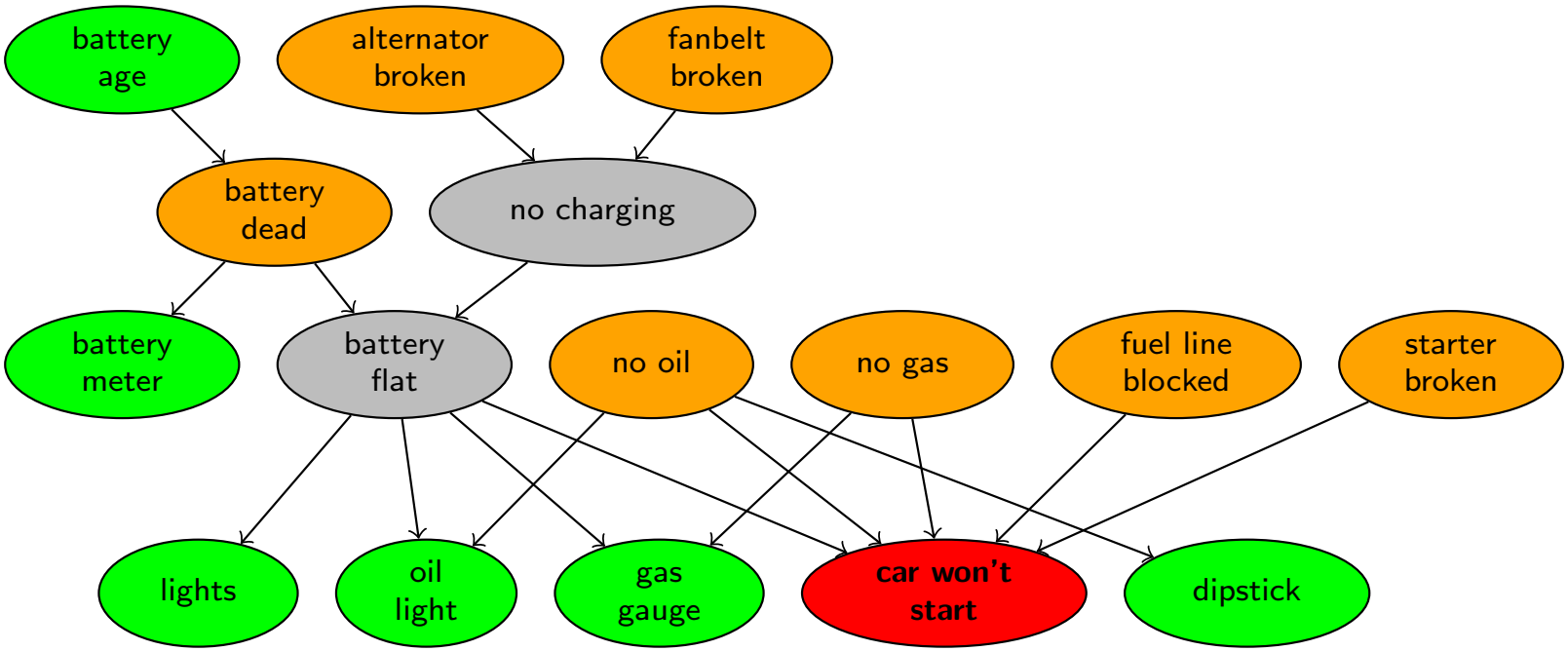
- **Observed symptom (red):** the initial evidence is *CarWon′tStart*





Example: Car diagnosis

- **Observed symptom (red):** the initial evidence is *CarWon′tStart*
- **Testable variables (green):** quantities that can be checked directly
- **Actionable fault or condition variables (orange):** possible causes for which a repair or corrective action may be available
- **Hidden mechanism variables (gray):** intermediate conditions not directly observed but can be inferred from the available evidence.



Learning Bayesian-network structure from data I

Structure learning

When domain knowledge alone does not determine the graph, we can use data to learn a plausible network structure: nodes, directed edges, and conditional independence assumptions.

Constraint-based methods

1. perform statistical tests of conditional independence;
2. infer which edges must be absent or may be present;
3. search for a graph compatible with these constraints.

Learning Bayesian-network structure from data II

Score-based methods

1. define a score that measures how well a candidate network explains the data, while penalizing unnecessary complexity;
2. search for the structure with the best score.

Important limitation

Different network structures may fit observational data equally well. **Data alone** therefore do not generally determine the causal direction of every edge.

Domain knowledge can provide constraints, such as temporal order or forbidden and required causal relations.

Once a structure has been selected, its conditional probability tables can be estimated from data or elicited from experts.

From Bayesian networks to causal networks

What we have seen

A Bayesian network represents a joint probability distribution through local conditional distributions and conditional independence assumptions.

However, observational data may support several network structures with different edge directions.

The causal question

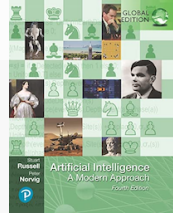
When can an edge $X \to Y$ be interpreted as saying that changing X would change Y, rather than merely that X and Y are probabilistically related?

Causal networks

A causal network enriches a Bayesian network with **causal assumptions**. These assumptions support reasoning about interventions, not only observations.

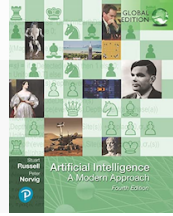
Causal networks

Causal networks

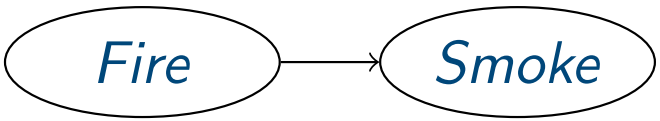


- In principle, any ordering of nodes permits a consistent construction of the network
- However, best to respect order of causality for many reasons, including compactness
- **Causal networks** are a restricted class of Bayesian networks that forbids all but causally compatible orderings

Causal networks

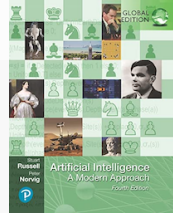


- Consider the following example:

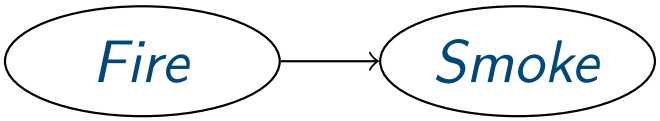


(a)

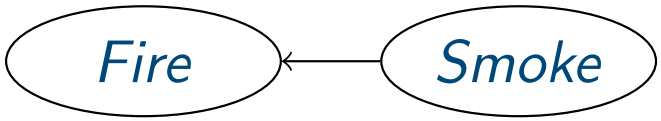
Causal networks



- Consider the following example:



(a)



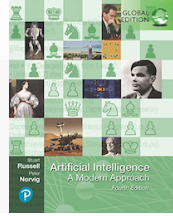
(b)

Both graphs can represent exactly the same joint distribution:

$P(Fire, Smoke) = P(Fire)\,P(Smoke \mid Fire) = P(Smoke)\,P(Fire \mid Smoke)$.

Observational data alone cannot distinguish the two directions.

- But *are these networks equivalent?*



Beyond probabilistic dependence: assignment I

Causal direction

A causal arrow $X_j \to X_i$ states that X_j is a direct input to the mechanism that generates X_i.

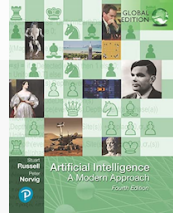
Example: fire and smoke

We draw: $Fire \longrightarrow Smoke$ because whether there is smoke directly depends, among other factors, on whether there is a fire:

$$Smoke = f_{Smoke}(Fire, U_{Smoke}).$$

Here, U_{Smoke} represents other factors that may also affect smoke, such as the amount of material burning or ventilation.

We do not draw $Smoke \to Fire$, because smoke is not an input to the mechanism that determines whether a fire starts: $Fire = f_{Fire}(U_{Fire})$.



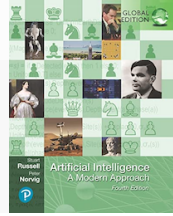
Beyond probabilistic dependence: assignment II

Structural equation

For each variable X_i, a structural equation specifies the mechanism that produces its value:

$$X_i = f_i\big(Parents(X_i), U_i\big).$$

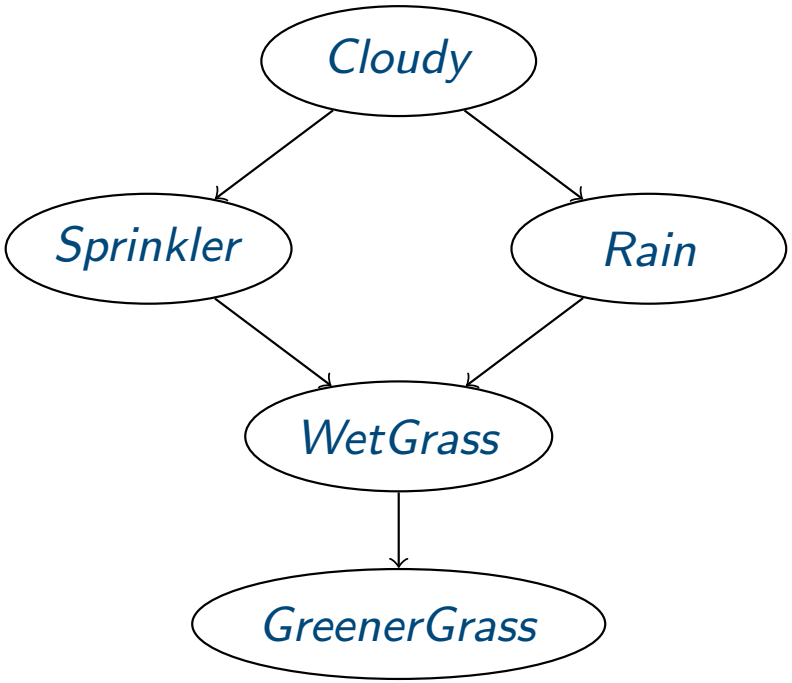
An arrow $X_j \to X_i$ is present exactly when X_j is a direct argument of the mechanism f_i.

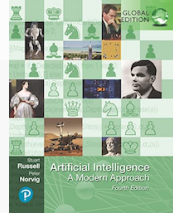


Lawn example

Invariance of structural equations

Structural equation represents an autonomous causal mechanism assumed to **remain unchanged** to measurements and local changes in the environment.

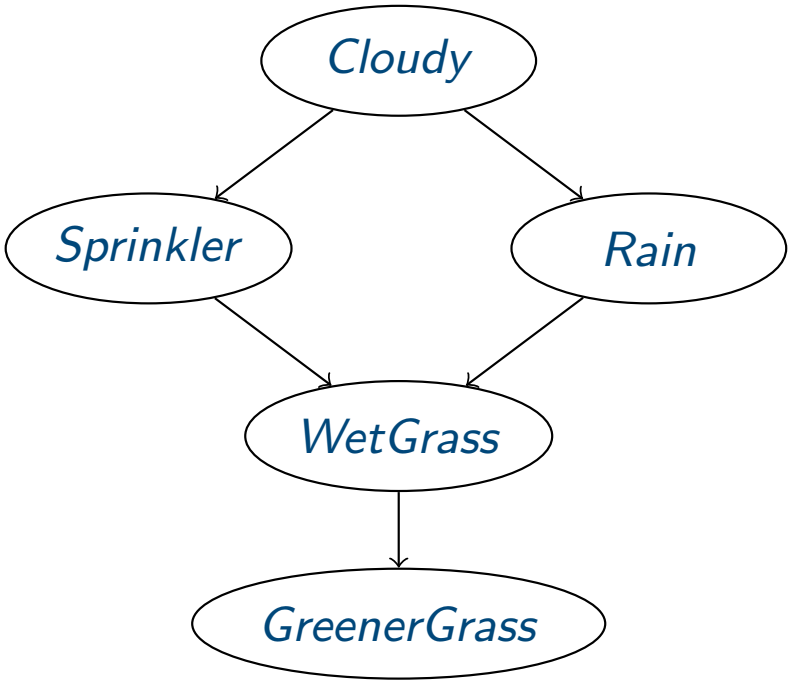


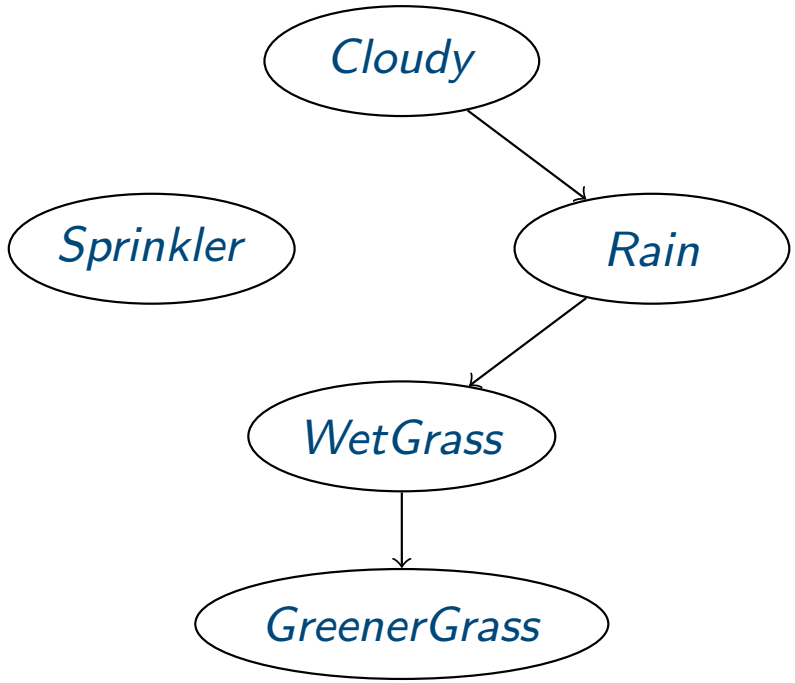


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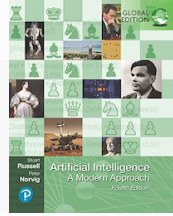
Invariance of structural equations

Structural equation represents an autonomous causal mechanism assumed to **remain unchanged** to measurements and local changes in the environment.





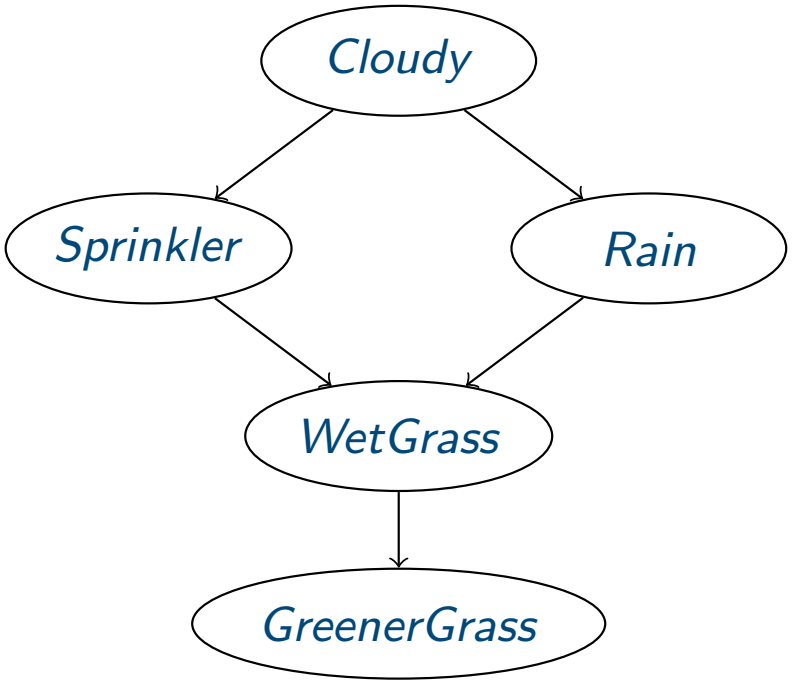
(disabled sprinkler)

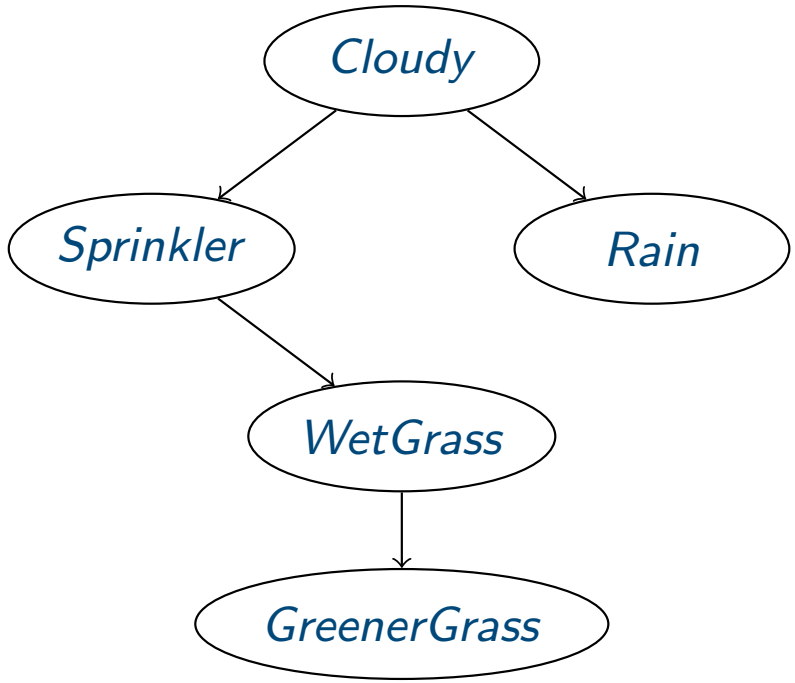


Lawn example

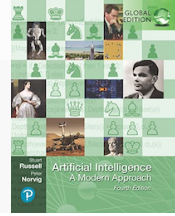
Invariance of structural equations

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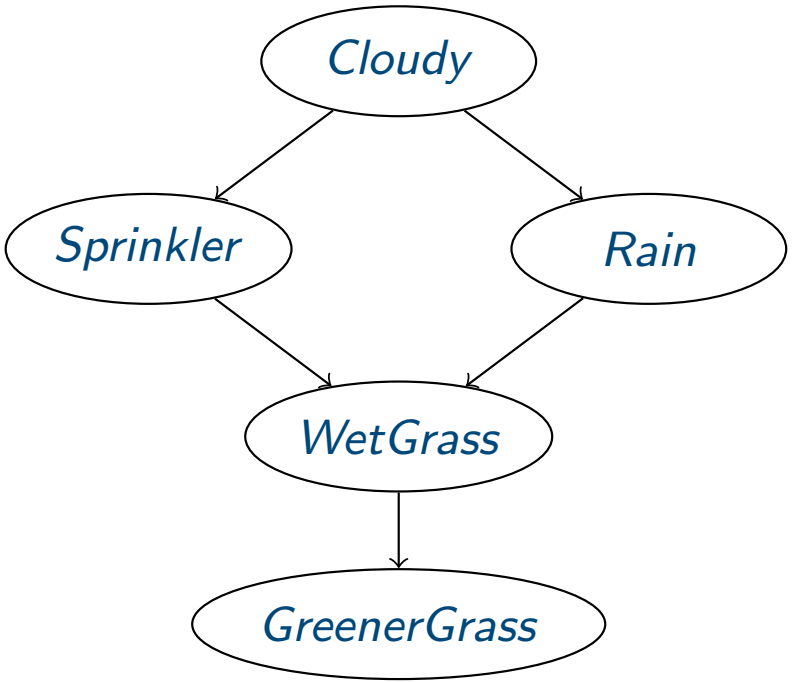
(lawn covered by a tent)



Stability for interventions

Stability of causal mechanisms

Structural equations represent causal mechanism that remains **stable** when we make observations or when other parts of the environment change.

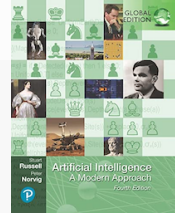


- Semantics of Bayes nets:

$$P(c, r, s, w, g)$$

- System of structural equations with "U-variables" (**unmodeled variables** or **error terms**):

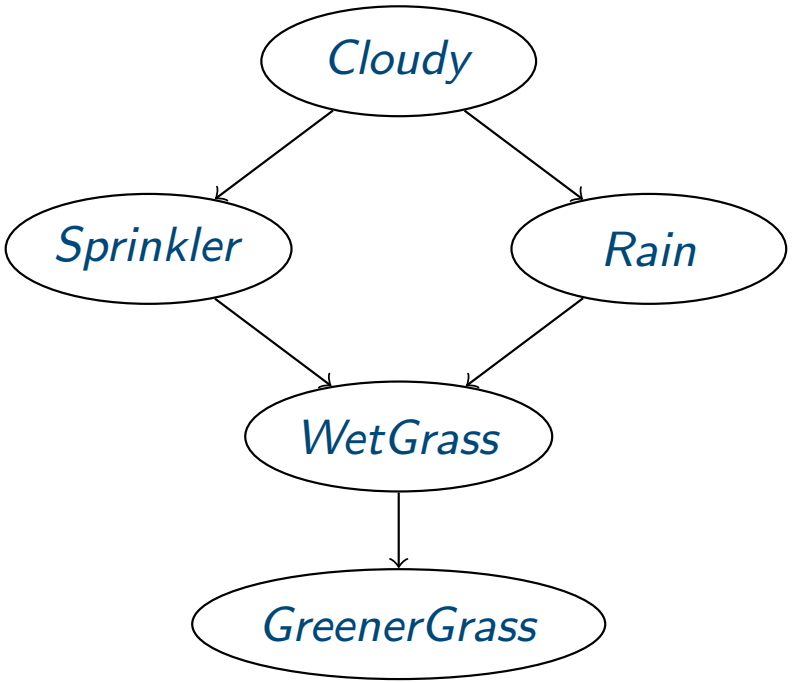
$$C = f_C(U_C)$$
$$R = f_R(C, U_R)$$
$$S = f_S(C, U_S)$$
$$W = f_W(S, R, U_W)$$
$$G = f_G(W, U_G)$$



Stability for interventions

Stability of causal mechanisms

Structural equations represent causal mechanism that remains **stable** when we make observations or when other parts of the environment change.



- Semantics of Bayes nets:

$$P(c, r, w, g|S = true)$$

- System of structural equations with "U-variables" (**unmodeled variables** or **error terms**):

$$C = f_C(U_C)$$
$$R = f_R(C, U_R)$$
$$S = do(S = true)$$
$$W = f_W(S, R, U_W)$$
$$G = f_G(W, U_G)$$

Observations and interventions I

A new kind of question

A Bayesian network lets us update beliefs after observing evidence.

A causal model also lets us ask what would happen if we actively changed a variable in the system.

Observations and interventions II

The lawn example

Consider two different questions:

$$P(WetGrass \mid Sprinkler = true)$$

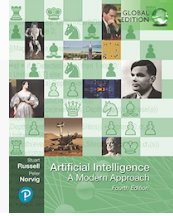
$$P(WetGrass \mid do(Sprinkler = true)).$$

The first asks: "What do we believe about the wet grass after observing that the sprinkler is on?"

The second asks: "What would happen to the wet grass if we turned the sprinkler on?"

Why can the answers differ?

Observing that the sprinkler is on may provide information about its usual causes, such as *Cloudy*. Turning it on externally breaks this usual causal mechanism.



The *do*-operator I

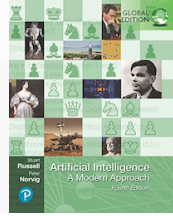
Intervention

The expression $do(X = x)$ denotes an intervention that sets X to the value x from outside the system.

If the original structural equation is $X = f_X(Parents(X), U_X)$, then the intervention replaces it with: $X = x$.

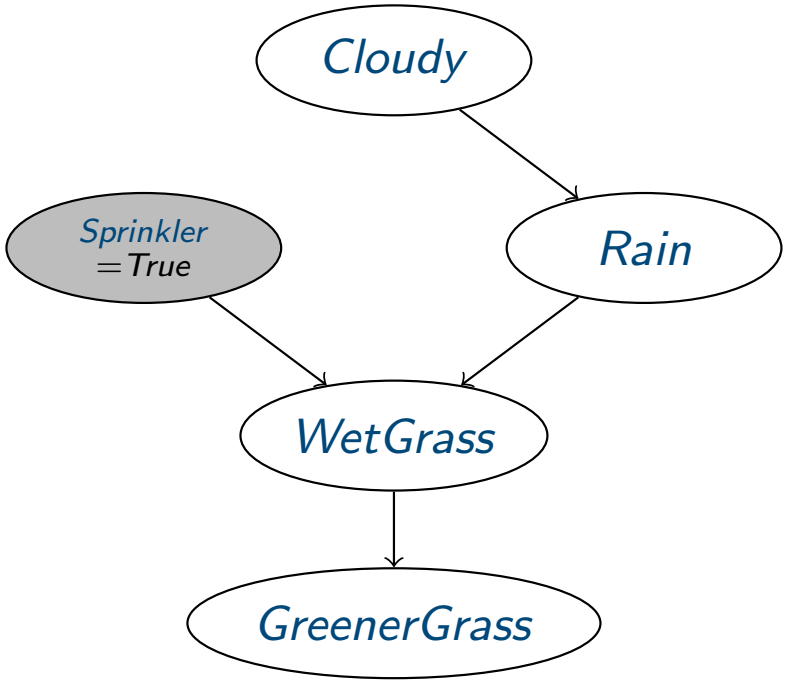
Graphical interpretation

An intervention removes all arrows entering X, because its value no longer depends on its usual causes. Arrows leaving X remain: X can still affect its usual consequences.



The *do*-operator II

- Stability is important for representing **interventions** and predicting their observable consequences



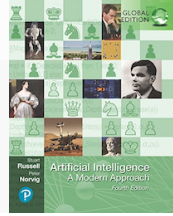
$do(Sprinkler = True)$:
we turn the sprinkler on

- Semantics of Bayes nets:

$$P(c, r, s, w, g)$$

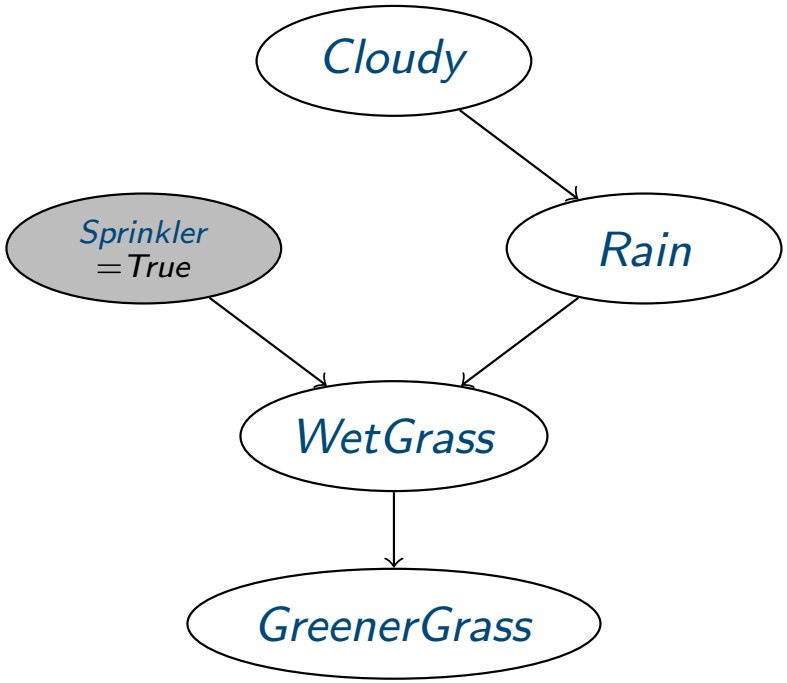
- System of structural equations with "U-variables" (**unmodeled variables** or **error terms**):

$$C = f_C(U_C)$$
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$$W = f_W(S, R, U_W)$$
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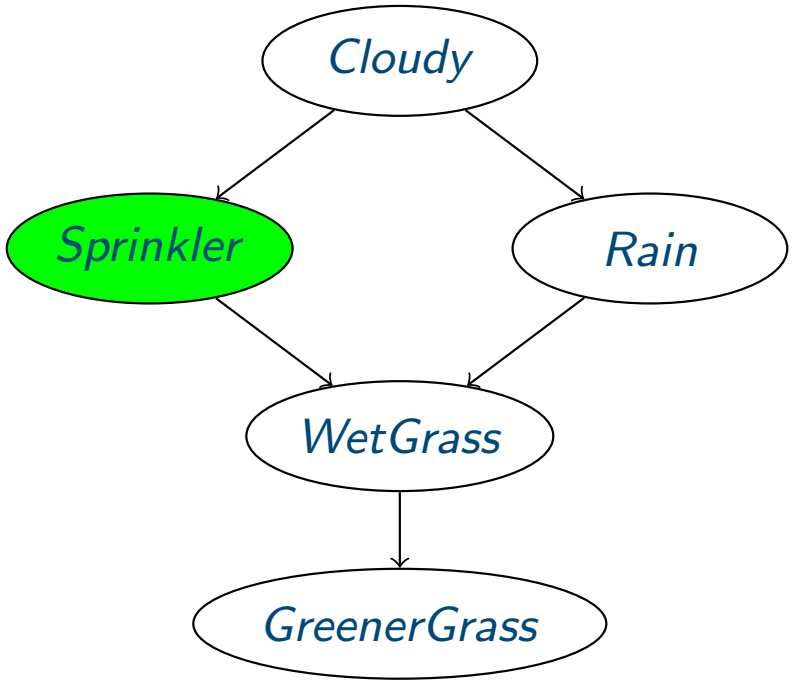


The *do*-operator

- What difference?



$P(WetGrass|do(Sprinkler = True))$:
we **turn** the sprinkler on



$P(WetGrass|Sprinkler = True)$:
we **observe** that sprinkler is on

Constructing Bayesian networks

Remark: choosing an edge direction

Remark: choosing an edge direction

Since *MaryCalls* and *JohnCalls* are dependent, the construction needs an edge between them. With the chosen ordering, we obtain:

$$MaryCalls \rightarrow JohnCalls.$$

With the reverse ordering, we would obtain:

$$JohnCalls \rightarrow MaryCalls.$$

Both choices are valid Bayesian-network factorizations, since:

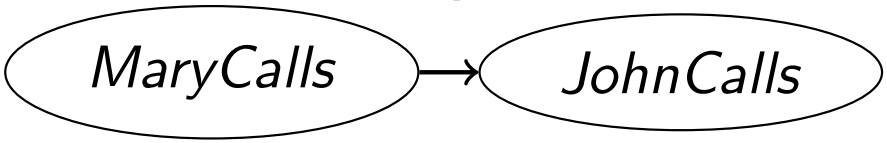
$$P(M, J) = P(M)P(J \mid M) = P(J)P(M \mid J).$$

The two arrows must not be included simultaneously, because they would form a directed cycle.

Two equivalent factorizations I

MaryCalls and *JohnCalls* dependence can be represented equivalently:

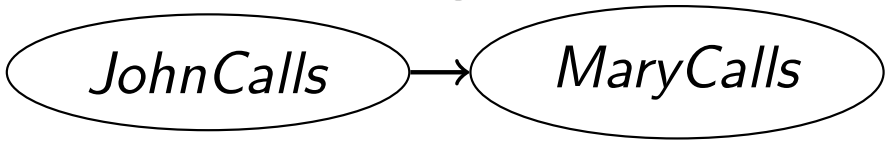
Ordering: M, J



$$P(M, J) = P(M)\,P(J \mid M).$$

Local distributions:
$P(M), \quad P(J \mid M).$

Ordering: J, M



$$P(M, J) = P(J)\,P(M \mid J).$$

Local distributions:
$P(J), \quad P(M \mid J).$

The two networks can represent exactly the same joint distribution

$P(M)P(J \mid M) = P(J)P(M \mid J)$.
They use different local conditional probability tables, but encode the same statistical dependence.

Two equivalent factorizations II

How do we choose an ordering and parent sets?

No causal knowledge: seek

- an ordering
- parent sets

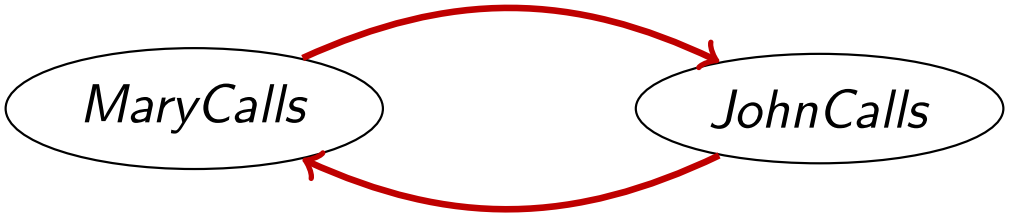
that exploit conditional independences in the domain and make the local conditional distributions easy to assess.

With causal knowledge:

- use it for selecting the right order

that often exposes conditional independences and therefore leads to a more compact network.

Why not use arrows in both directions? I



Not a Bayesian network

There is no ordering in which one variable can be specified before the other:

$$P(M \mid J), \qquad P(J \mid M).$$

The product of these two distributions does **not** define the joint distribution:

$$P(M, J) \neq P(M \mid J)\, P(J \mid M).$$

A Bayesian network needs a non-circular factorization

Start from root variables and then specify each variable given variables that come earlier in the ordering.

What are we trying to achieve? I

Our goal

We want to **model an uncertain domain** in order to assess probabilities of events and update them when new evidence becomes available.

For a domain described by variables $X_1, \ldots, X_n$, this would require, in principle, a full joint distribution:

$$P(X_1, \ldots, X_n).$$

The challenge

A **full joint distribution grows exponentially** with the number of variables. We therefore seek a compact representation that preserves the relevant dependencies in the domain.

What are we trying to achieve? II

The Bayesian-network idea

Choose an **ordering** of the variables and, for each variable X_i, identify a **small set of earlier variables** that directly affect its probability:

$$P(X_i \mid Parents(X_i)).$$

These local conditional distributions represent the full joint distribution:

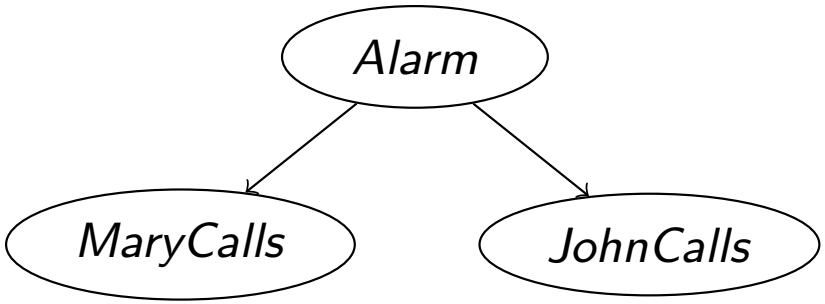
$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid Parents(X_i)).$$

We are not trying to make every variable independent. We are trying to **find an ordering and a small set of direct parents for each variable**, so that the remaining earlier variables do not add further information once those parents are known.

Why a good ordering matters

Suppose that domain knowledge tells us: $MaryCalls \perp JohnCalls \mid Alarm$.

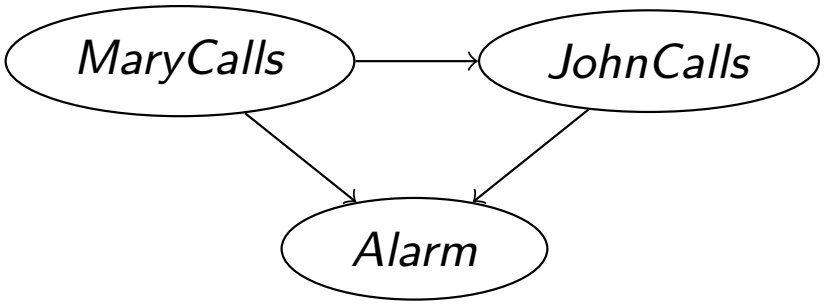
Causal ordering



$$P(A, M, J) = P(A)P(M \mid A)P(J \mid A).$$

How many free parameters?

Arbitrary ordering: M, J, A



$$P(A, M, J) = P(M)P(J \mid M)P(A \mid M, J).$$

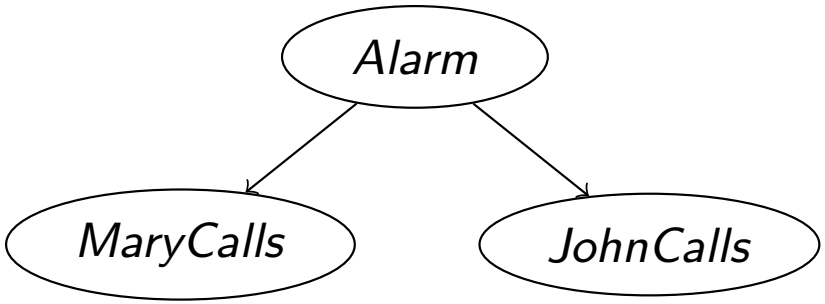
How many free parameters?

Both networks represent same probability distribution. However, the causal ordering makes the conditional independence explicit and requires fewer parameters.

Why a good ordering matters

Suppose that domain knowledge tells us: $MaryCalls \perp JohnCalls \mid Alarm$.

Causal ordering

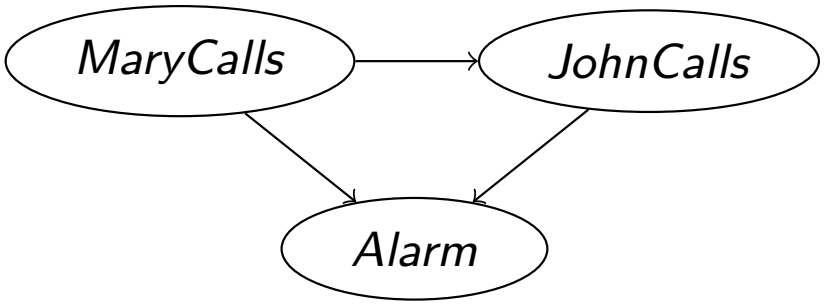


$$P(A, M, J) = P(A)P(M \mid A)P(J \mid A).$$

$$\underbrace{1}_{P(A)} + \underbrace{2}_{P(M|A)} + \underbrace{2}_{P(J|A)} = 5$$

free parameters.

Arbitrary ordering: M, J, A



$$P(A, M, J) = P(M)P(J \mid M)P(A \mid M, J).$$

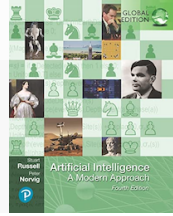
$$\underbrace{1}_{P(M)} + \underbrace{2}_{P(J|M)} + \underbrace{4}_{P(A|M,J)} = 7$$

free parameters.

Both networks represent same probability distribution. However, the causal ordering makes the conditional independence explicit and requires fewer parameters.

Representing conditional distributions

Compact local models beyond conditional probability tables



Beyond conditional probability tables

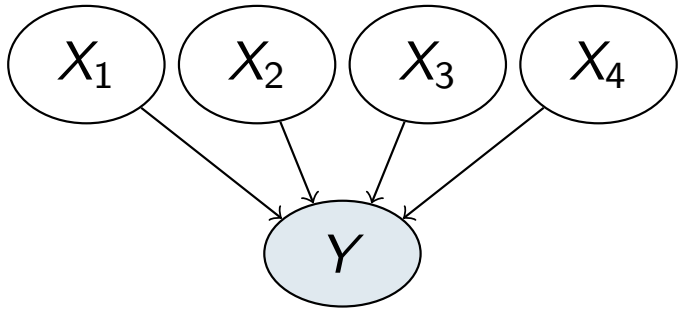
Conditional probability tables

Lists one probability distribution for every combination of parent values:

- CPT grows exponentially with number of parents
- CPT becomes infinite with continuous-valued parent or child

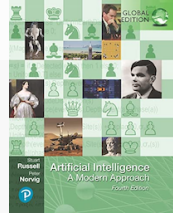
- n Boolean parents: CPT has 2^n rows
- Continuous parent: infinitely possible values
- Continuous child: needs conditional density

CPTs are practical only when a node has few discrete parents.



X_1	X_2	X_3	X_4	$P(Y = true \mid X_1, \ldots, X_4)$
T	T	T	T	p_1
T	T	T	F	p_2
⋮	⋮	⋮	⋮	⋮
F	F	F	F	p_{16}

$2^4 = 16$ rows already; in general, 2^n.



Solution: Compact conditional distributions

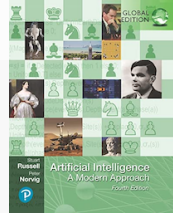
Canonical conditional distribution

Specifies $P(X \mid Parents(X))$ through a **compact functional form** with a **small number of parameters** rather than listing a separate table entry for every possible combination of parent values.

The key idea

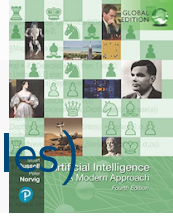
Replace a large CPT with a compact model.

A CPT is the general representation; canonical distributions are compact representations tailored to recurrent structures in the domain.

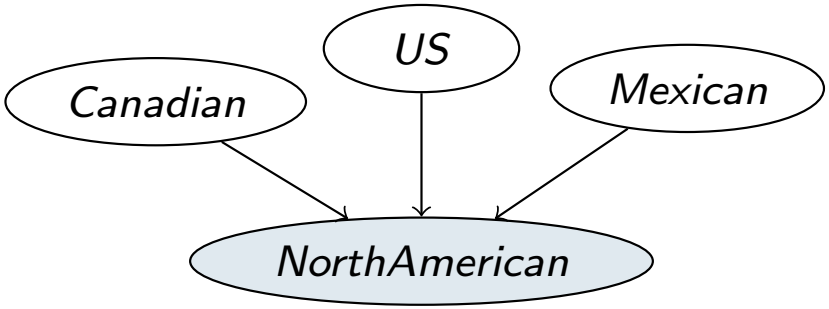


Examples: Compact conditional distributions

- Deterministic nodes: the child is an exact function of its parents;
- Noisy-OR: many Boolean causes independently trigger the same Boolean effect;
- Hybrid models: involving continuous/non continuous variables;
- . . . other compact, domain-specific parametric forms.



Deterministic nodes: a rule replaces a CPT (Boolean variables)



A local fragment of a larger Bayesian network.

$NorthAmerican \equiv Canadian \vee US \vee Mexican.$

C	U	M	$P(N = true \mid C, U, M)$
F	F	F	0
F	F	T	1
F	T	F	1
F	T	T	1
T	F	F	1
T	F	T	1
T	T	F	1
T	T	T	1

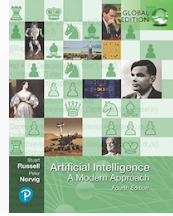
one logical rule $\iff$ $2^3 = 8$ CPT rows.

What the rule represents

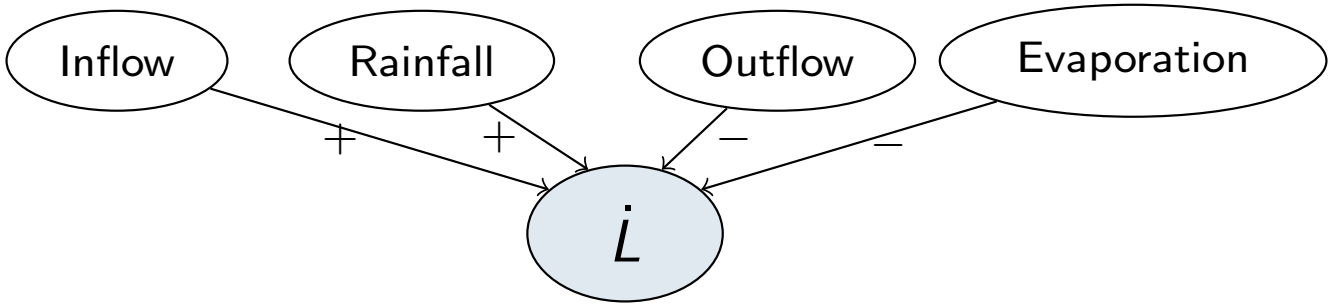
The rule compactly specifies the complete conditional distribution $P(NorthAmerican \mid Canadian, US, Mexican)$.

$$P(NorthAmerican = true \mid C = c, U = u, M = m) = \begin{cases} 0, & \text{if } c = u = m = false, \\ 1, & \text{otherwise.} \end{cases}$$

With n Boolean parents, a CPT has 2^n rows. The rule: size **linear in the number of parents** and does not require estimating any probability parameters.



Deterministic nodes (Continuous variables)



$\dot{L}$: rate of change of the water level

A continuous deterministic relationship

Let L denote the water level. Its rate of change is determined by the four input variables:

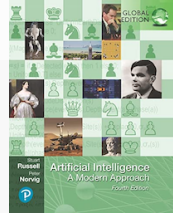
$$\dot{L} = \mathit{Inflow} + \mathit{Rainfall} - \mathit{Outflow} - \mathit{Evaporation}.$$

For example, if $\mathit{Inflow} = 5$, $\mathit{Rainfall} = 2$, $\mathit{Outflow} = 3$, $\mathit{Evaporation} = 1$, then: $\dot{L} = 5 + 2 - 3 - 1 = 3$. Therefore:

$$P(\dot{L} = 3 \mid I = 5, R = 2, O = 3, E = 1) = 1,$$

$$P(\dot{L} = x \mid I = 5, R = 2, O = 3, E = 1) = 0 \qquad \text{for every } x \neq 3.$$

The conditional distribution is thus entirely concentrated on the value of the equation.



Deterministic nodes: definitions

Deterministic node

A node X is deterministic if its value is completely determined by the values of its parents:

$$X = f(Parents(X)).$$

Equivalently, once $Parents(X) = pa$ is known, the conditional distribution of X assigns probability 1 to the value $f(pa)$.

From deterministic OR to Noisy-OR

When an OR rule is too strong

Suppose that *Flu*, *Allergy*, and *Cold* may each cause *Sneezing*.

$$Sneezing \equiv Flu \vee Allergy \vee Cold$$

would mean that every active cause produces sneezing with certainty.

In practice, this is rarely true:

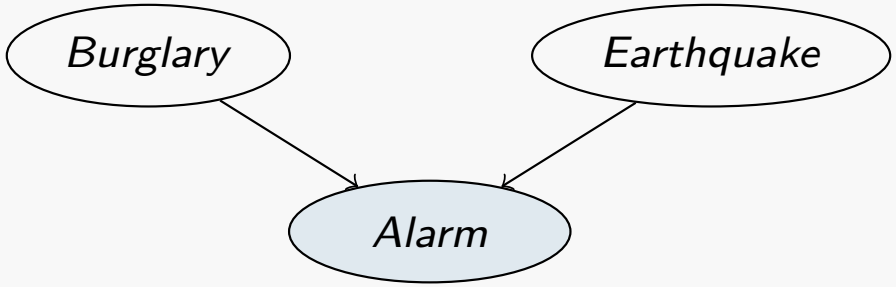
- a person with flu does not necessarily sneeze;
- an allergy may cause sneezing more often than a cold;
- several possible causes may be present at the same time.

Noisy-OR: idea

We need a compact conditional distribution in which each possible cause can independently make the effect more likely, without making it certain.

From a logical OR to an uncertain causal relation

A burglary and an earthquake can both trigger an alarm.



A deterministic model would state: $Alarm \equiv Burglary \vee Earthquake$.
$\Rightarrow$ whenever burglary and no earthquake occur, alarm must sound:

$$P(Alarm = true \mid Burglary = true, Earthquake = false) = 1.$$

Why is this unrealistic?

A burglary can occur without triggering the alarm: the system may be disarmed, a sensor may fail, or the alarm may malfunction.

Thus, a cause can make an effect **more likely** without determining it:

$$P(Alarm = true \mid Burglary = true, Earthquake = false) = 0.95.$$

Noisy-OR

Noisy-OR

A compact conditional distribution for a Boolean effect Y with Boolean causes $X_1, \ldots, X_n$. Each active cause X_i has a causal mechanism that can trigger Y, but may fail with inhibition probability q_i. Conditional on the active causes, the **failures** of these mechanisms are **assumed to be independent**. The effect is false only if every active cause fails:

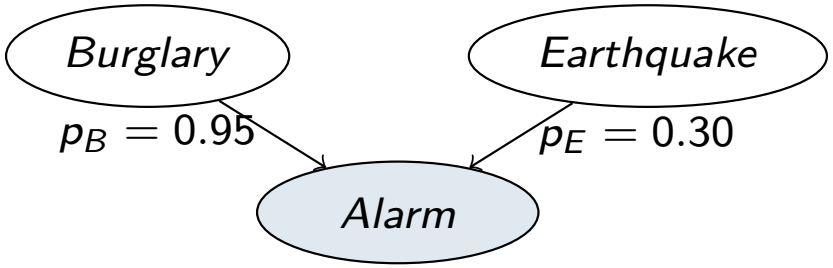
$$P(Y = \mathit{false} \mid X_1, \ldots, X_n) = \prod_{i:X_i=\mathit{true}} q_i.$$

Therefore: $P(Y = \mathit{true} \mid X_1, \ldots, X_n) = 1 - \prod_{i:X_i=\mathit{true}} q_i$.

Note: the basic Noisy-OR assumption

If none of the explicitly modelled causes is active, the effect cannot occur:
$P(Y = \mathit{true} \mid X_1 = \mathit{false}, \ldots, X_n = \mathit{false}) = 0$.
$\Rightarrow$ address this limitation by introducing a leak node: additional **always-active parent** that represents all unmodelled causes.

Example: A Noisy-OR model I



Link probabilities

$$p_B = P(Alarm = true \mid Burglary = true,\ Earthquake = false) = 0.95,$$
$$p_E = P(Alarm = true \mid Burglary = false,\ Earthquake = true) = 0.30.$$

Thus, the corresponding inhibition probabilities are:

$$q_B = 1 - p_B = 0.05, \qquad q_E = 1 - p_E = 0.70.$$

When both causes are active

The alarm is absent only if both causal mechanisms fail:

$P(Alarm = false \mid Burglary = true,\ Earthquake = true)$

$$= q_B q_E = 0.05 \cdot 0.70 = 0.035.$$

Therefore: $P(Alarm = true \mid Burglary = true,\ Earthquake = true) = 1 - 0.035 = 0.965.$

Example: A Noisy-OR model II

The conditional probability table implied by the Noisy-OR

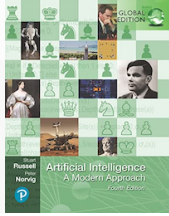
Burglary	Earthquake	$P(Alarm = true \mid B, E)$
false	false	0
true	false	$p_B = 0.95$
false	true	$p_E = 0.30$
true	true	$1 - (1 - p_B)(1 - p_E) = 0.965$

Compact representation

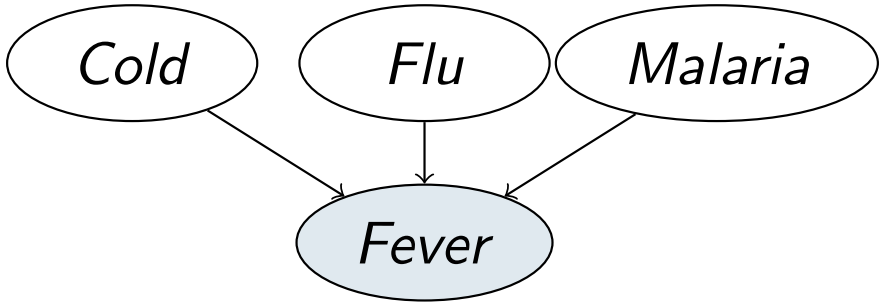
Rather than specifying a separate probability for every parent configuration, the Noisy-OR specifies the whole CPT using only:

$$p_B = 0.95, \qquad p_E = 0.30.$$

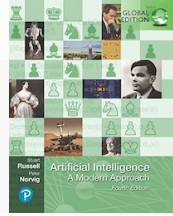
More generally, **for n Boolean causes**, it uses **n link probabilities** rather than a table with 2^n rows.



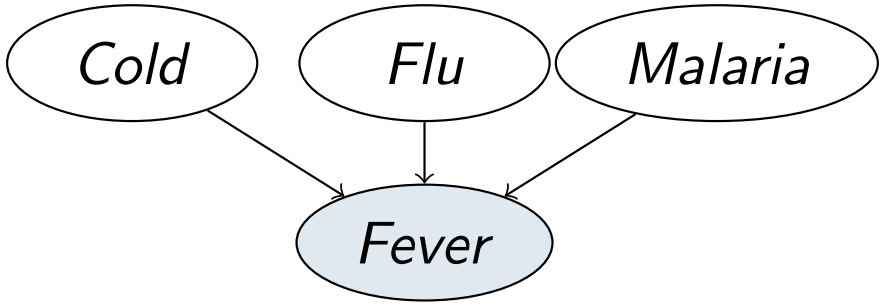
Additional example: A Noisy-OR model



Cold	*Flu*	*Malaria*	$P(Fever)$	$P(\neg Fever)$
F	F	F	0.0	
F	F	T		**0.1**
F	T	F		**0.2**
F	T	T		
T	F	F		**0.6**
T	F	T		
T	T	F		
T	T	T		

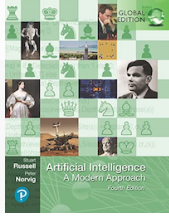


Additional example: A Noisy-OR model

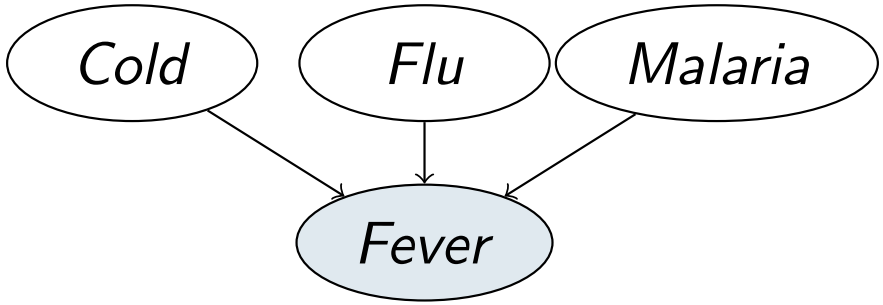


Cold	*Flu*	*Malaria*	$P(Fever)$	$P(\neg Fever)$
F	F	F	0.0	
F	F	T		**0.1**
F	T	F		**0.2**
F	T	T		
T	F	F		**0.6**
T	F	T		
T	T	F		
T	T	T		

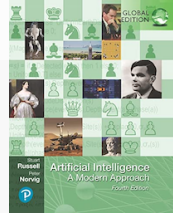
Number of parameters **linear** in number of parents



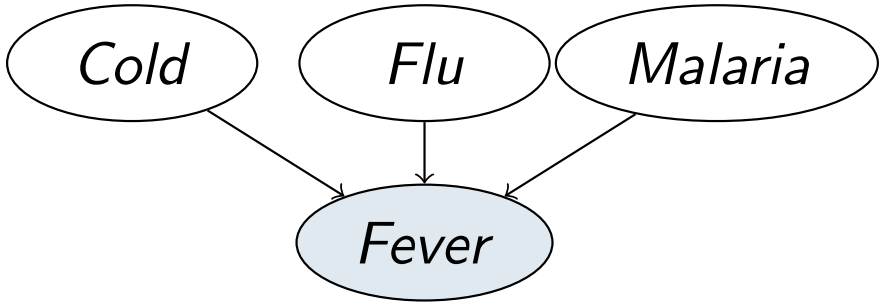
Additional example: A Noisy-OR model



Cold	*Flu*	*Malaria*	$P(Fever)$	$P(\neg Fever)$
F	F	F	0.0	
F	F	T		**0.1**
F	T	F		**0.2**
F	T	T		$0.02 = 0.2 \times 0.1$
T	F	F		**0.6**
T	F	T		$0.06 = 0.6 \times 0.1$
T	T	F		$0.12 = 0.6 \times 0.2$
T	T	T		$0.012 = 0.6 \times 0.2 \times 0.1$



Additional example: A Noisy-OR model



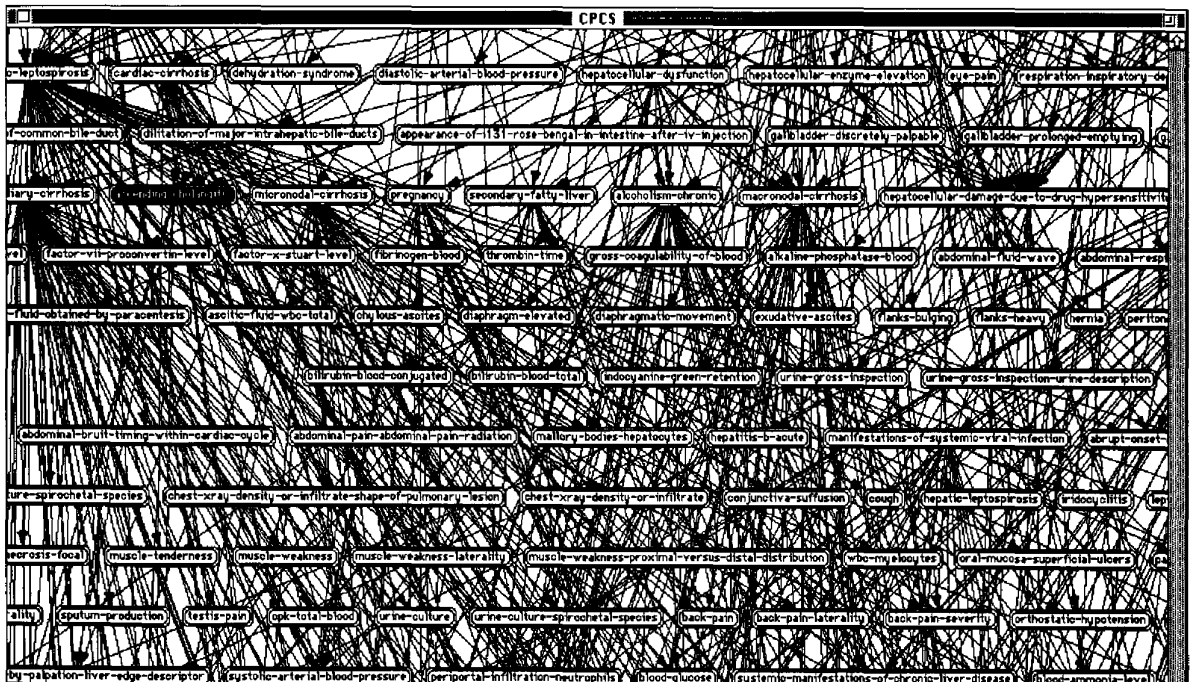
Cold	*Flu*	*Malaria*	$P(Fever)$	$P(\neg Fever)$
F	F	F	0.0	1.0
F	F	T	0.9	**0.1**
F	T	F	0.8	**0.2**
F	T	T	0.98	$0.02 = 0.2 \times 0.1$
T	F	F	0.4	**0.6**
T	F	T	0.94	$0.06 = 0.6 \times 0.1$
T	T	F	0.88	$0.12 = 0.6 \times 0.2$
T	T	T	0.988	$0.012 = 0.6 \times 0.2 \times 0.1$

A Bayesian network for internal medicine

Knowledge Engineering for Large Belief Networks, Pradhan et al., UAI 1994

448 nodes 908 arcs.

Modelled via Noisy-MAX (Noisy-OR generalization), together with leak probabilities.

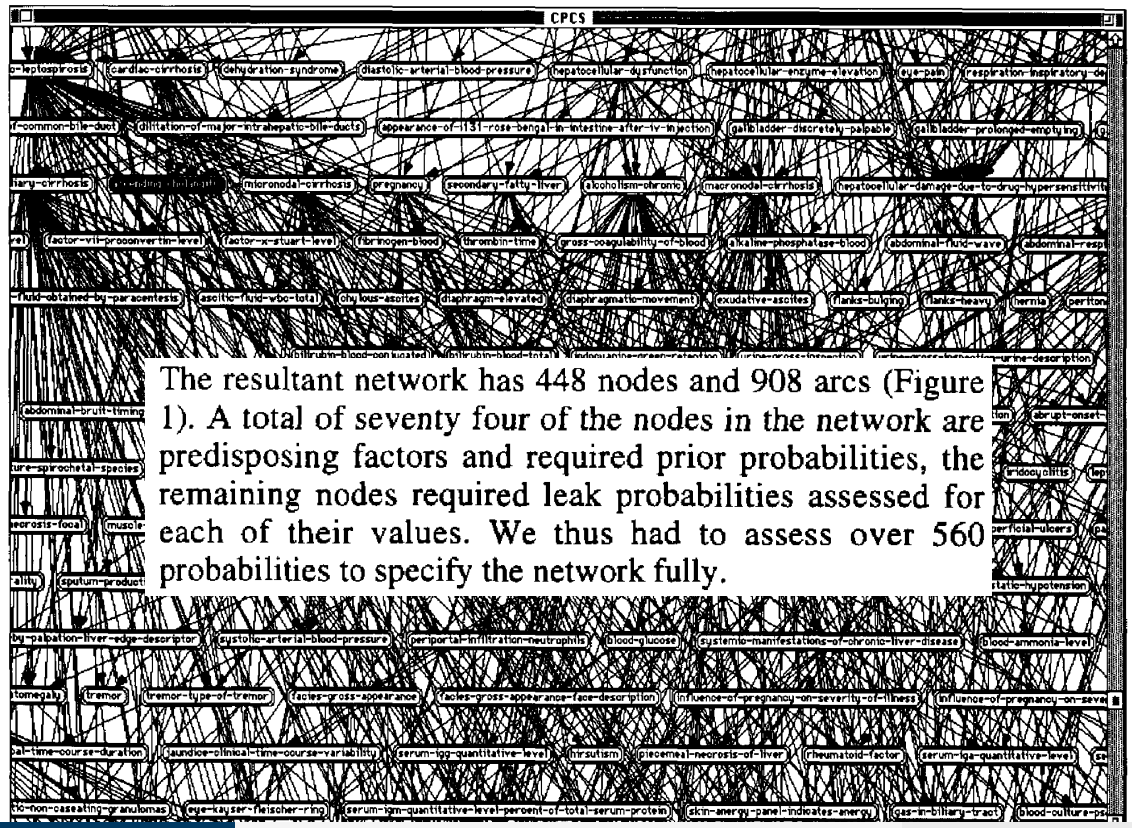


A Bayesian network for internal medicine

Knowledge Engineering for Large Belief Networks, Pradhan et al., UAI 1994

448 nodes 908 arcs.

Modelled via Noisy-MAX (Noisy-OR generalization), together with leak probabilities.



A third family of compact conditional distribution

So far, we have seen two compact conditional distributions:

- deterministic functions, which determine the child value exactly;
- Noisy-OR, for uncertain causal relations between Boolean variables.

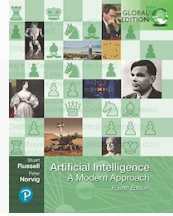
What if the child is continuous and non-deterministic—or, more generally, the network is hybrid?

Its local model is a **conditional probability distribution**:

- a conditional density when the child is continuous;
- a conditional probability model when the child is discrete.

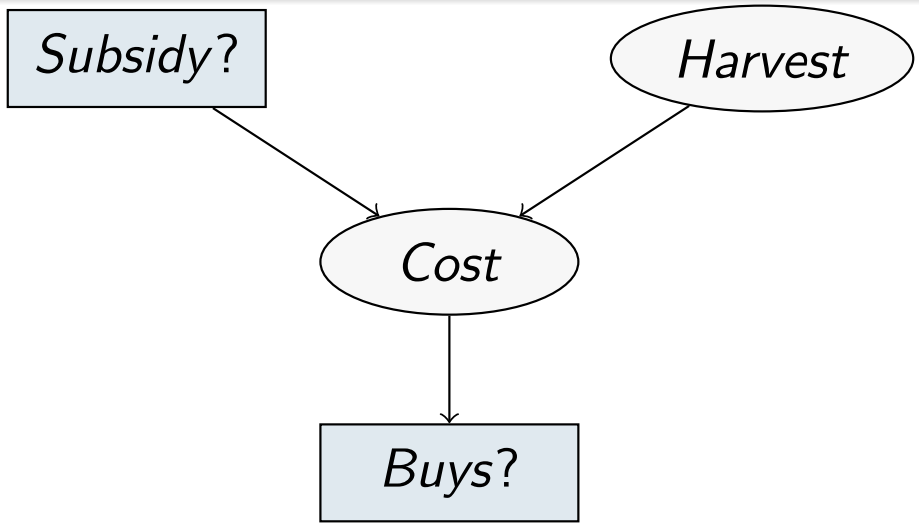
Hybrid Bayesian network

A Bayesian network containing **both discrete and continuous** random variables. Its local conditional models may therefore combine CPT, conditional densities, and other compact parametric distributions.



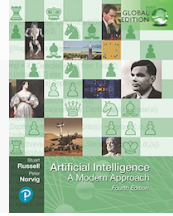
Hybrid network: a running example

Model how the cost of harvest depends on amount harvested and on availability of subsidy. The resulting cost influences whether a buyer decides to buy.



- discrete variables: *Subsidy*?, *Buys*?
- continuous variables: *Harvest*, *Cost*.

$$P(Cost \mid Harvest, Subsidy?) \quad \text{and} \quad P(Buys? \mid Cost).$$



Hybrid (discrete+continuous) networks

How can we model a hybrid network?

- Option 1: discretization
  Replace continuous variables with a finite set of intervals
  $\rightarrow$ use ordinary CPTs.
  Trade-off: few intervals may lose relevant information; many intervals can lead to large CPTs.

- Option 2: compact parametric conditional models
  Use a **canonical family** specified by a finite number of parameters, rather than by enumerating every possible parent configuration:
  - a continuous child with continuous and/or discrete parents (e.g., *Cost*);
  - a discrete child with one or more continuous parents (e.g., *Buys*?).



Continuous child with continuous and/or discrete parents I

Conditional density for a continuous child

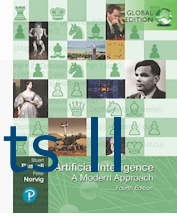
For a continuous child variable Y, the local conditional distribution is specified by a **conditional probability density**

$$p(y \mid Parents(Y)).$$

For every fixed assignment of the parent values, this density is nonnegative and integrates to 1 over the possible values of Y.

A compact canonical family represents this conditional density through a finite number of parameters.

- Separate density is defined for each assignment of discrete parents
- The density can then depend on the values of the continuous parents



Continuous child with continuous and/or discrete parents II

Linear-Gaussian model

The cost is modelled as a linear trend plus remaining uncertainty:

$$Cost = a_t\, Harvest + b_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_t^2).$$

Equivalently, for fixed values $Harvest = h$ and $Subsidy? = t$:

$$Cost \mid (Harvest = h,\ Subsidy? = t) \sim \mathcal{N}(a_t h + b_t, \sigma_t^2).$$

How to read this model

For a fixed harvest h and subsidy value t, the model does **not** predict one exact cost. It assigns a bell-shaped distribution of plausible cost values.

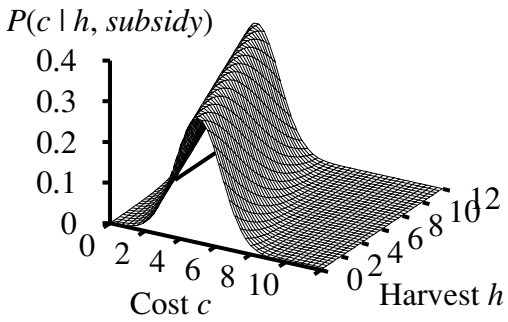
$$\mathbb{E}[Cost \mid Harvest = h,\ Subsidy? = t] = a_t h + b_t.$$

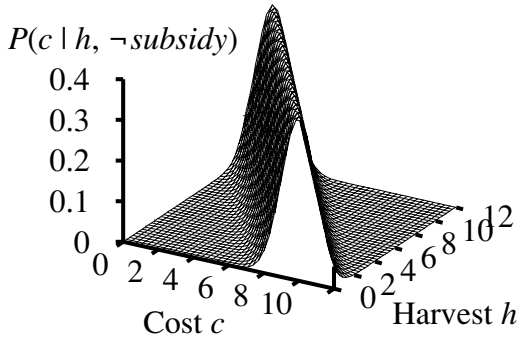
- a_t: how the expected cost changes as the harvest increases;
- b_t: the intercept of this linear relationship;
- σ_t^2: the remaining uncertainty around the expected cost.

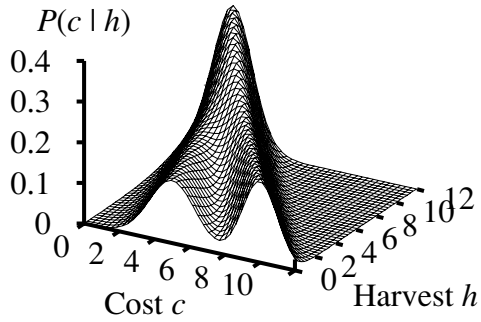
The parameters may differ for $Subsidy? = true$ and $Subsidy? = false$.



Continuous child with continuous and/or discrete parents III

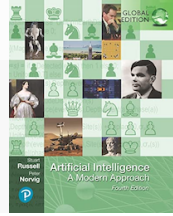






Linear Gaussian and conditional Gaussian networks

- If all variables continuous, each node has a linear Gaussian (LG) conditional distribution, the full joint distribution is a multivariate Gaussian: $P(X_1, \ldots, X_n) = \mathcal{N}(\boldsymbol{\mu}, \Sigma)$.
- If also discrete parents, conditional Gaussian network: for each assignment $\mathbf{D} = \mathbf{d}$ of the discrete variables, $P(\mathbf{X}_{\text{cont}} \mid \mathbf{D} = \mathbf{d})$ is a multivariate Gaussian.
- If discrete parents are unobserved, average over their possible values, e.g., $P(\mathit{Cost} \mid \mathit{Harvest}) = \sum_{t \in \{\mathit{true}, \mathit{false}\}} P(C \mid H,\ S? = t)\, P(S? = t \mid H)$. The resulting distribution is generally a **mixture of Gaussians**.



Discrete child with a continuous parent I

Conditional probability model for a discrete child

For a discrete child variable Y with continuous parent variables $\mathbf{X}$, the local model specifies a probability distribution over the possible values of Y for every parent value $\mathbf{x}$:

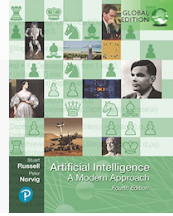
$$P(Y = y \mid \mathbf{X} = \mathbf{x}).$$

Since the parent domain is continuous, these probabilities cannot be listed in a finite CPT. A compact canonical family represents them through a function with a finite number of parameters.

A soft threshold

A deterministic rule might state: $Buys? = true \quad \text{iff} \quad Cost < \mu$.

In practice, decision to buy is uncertain: as the cost increases, buying becomes **gradually less likely**.



Discrete child with a continuous parent II

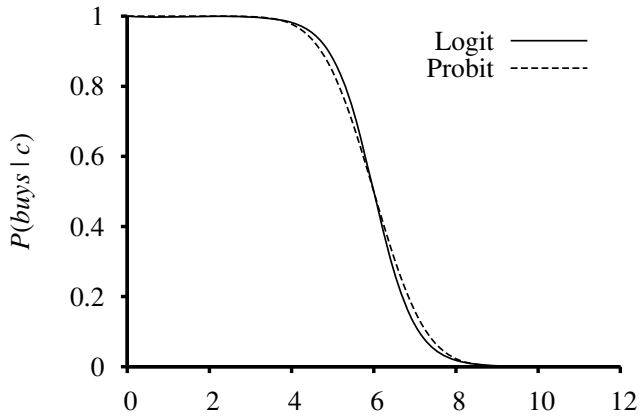
Logistic (sigmoid) model

$P(Buys? = true \mid Cost = c) = \frac{1}{1+exp(-2\frac{-c+\mu}{\sigma})}$

- μ cost at which prob. of buying is 0.5;
- s controls how gradual the transition is;
- c: prob of buying decreases.

Probit model

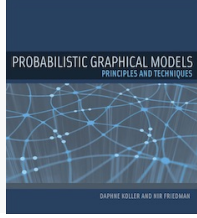
$P(Buys? = true \mid Cost = c) = \Phi\left(\frac{\mu - c}{s}\right)$, Φ cumulative distribution f of standard Gaussian.



Both curves implement a soft threshold. The logistic curve has slightly heavier tails than the probit curve.

Representing conditional distributions

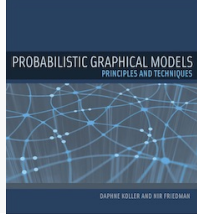
Bayesian networks over time



Dynamic Bayesian networks I

From a state to a trajectory

So far, a Bayesian network has described the variables of a system at one point in time. To reason about an **evolving world**, we must represent how the system state **changes over time**.



Dynamic Bayesian networks II

Dynamic Bayesian network (DBN)

A compact, **template-based** representation of a **probability distribution over** the evolution of a system through **time**.

System state and template variables

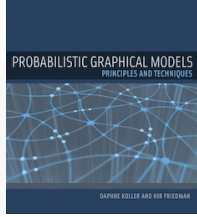
The system state at time t is a set of random variables:

$$\mathcal{X}^{(t)} = \{X_1^{(t)}, \ldots, X_n^{(t)}\}.$$

template variable X_i denotes the same quantity at every time step; $X_i^{(t)}$ is its instance at time t.

Trajectory

A trajectory is a complete sequence of system states: $\left(\mathcal{X}^{(0)}, \mathcal{X}^{(1)}, \ldots, \mathcal{X}^{(T)}\right)$.

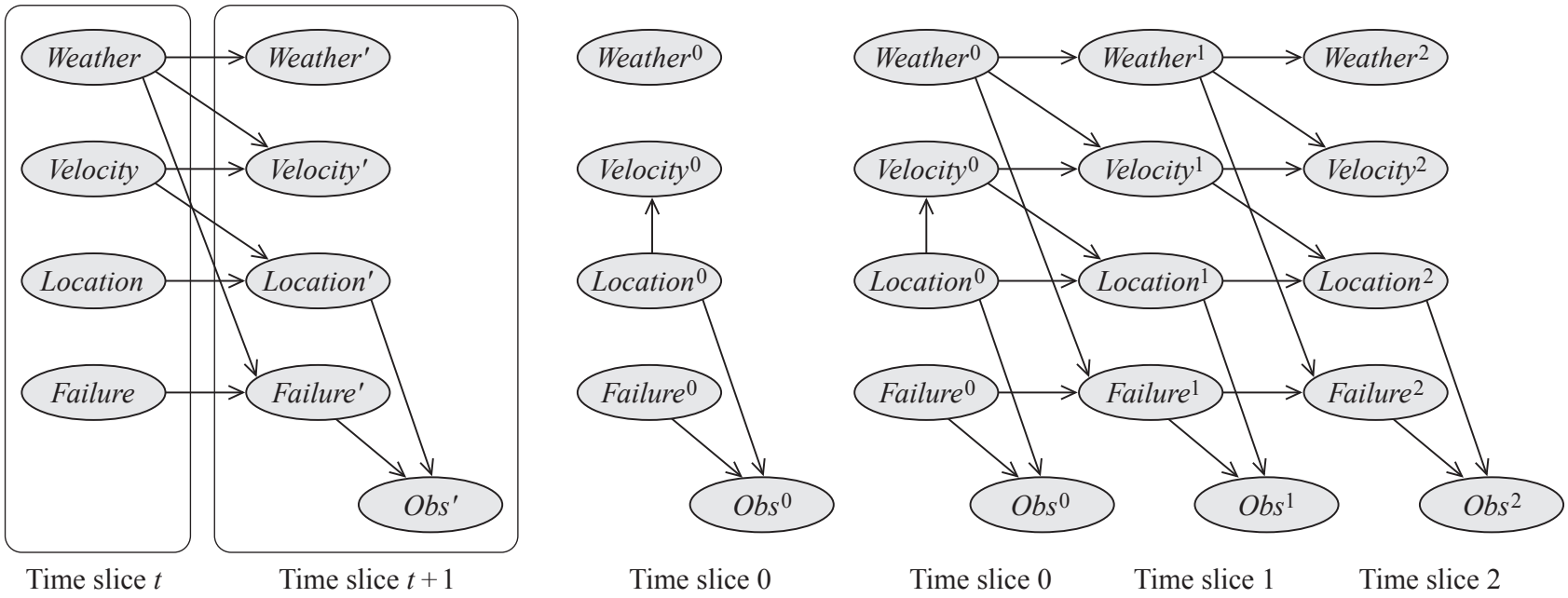


Dynamic Bayesian networks III

DBN

Specifies a joint probability distribution over trajectories:

$$P\left(\mathcal{X}^{(0)}, \mathcal{X}^{(1)}, \ldots, \mathcal{X}^{(T)}\right).$$



(a) $\mathcal{B}_{\rightarrow}$ (b) $\mathcal{B}_0$ (c) DBN unrolled over 3 steps

Example

Weather and traffic

Suppose that we monitor the weather and the traffic conditions every day:

$$\mathcal{X}^{(t)} = \{Weather^{(t)}, Traffic^{(t)}\}.$$

At each time step, the system state specifies the values of both variables.

A possible trajectory over three time steps

	$t=0$	$t=1$	$t=2$
Weather	sunny	rain	rain
Traffic	low	high	high

Another possible trajectory

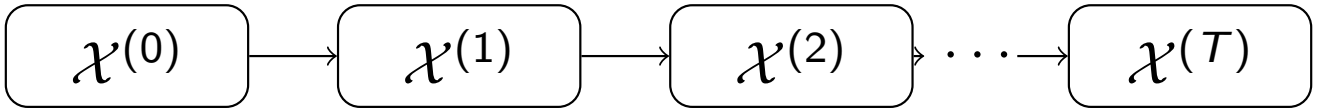
	$t=0$	$t=1$	$t=2$
Weather	sunny	sunny	cloudy
Traffic	low	low	medium

A dynamic probabilistic model assigns a probability to every possible trajectory, not only to the state at a single time step.

Template-based representations over time

Why use a template?

Many systems **evolve according to the same local mechanisms** at every time step. Rather than drawing and specifying a new Bayesian network for every instant, we specify a structure once and repeat it over time.



The two-slice transition template

First-order Markov assumption

The state at the next time step depends only on the current state:

$$P\left(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(0)}, \ldots, \mathcal{X}^{(t)}\right) = P\left(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(t)}\right).$$

Two-slice transition template

A two-slice template specifies dependencies between one time step t and the next one $t+1$. **Unrolled** repeatedly to represent a trajectory of arbitrary length:

$$\mathcal{X}^{(0)}, \mathcal{X}^{(1)}, \ldots, \mathcal{X}^{(T)}.$$

Distribution over a trajectory

A first-order DBN specifies:

$$P\left(\mathcal{X}^{(0)}, \ldots, \mathcal{X}^{(T)}\right) = P\left(\mathcal{X}^{(0)}\right) \prod_{t=0}^{T-1} P\left(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(t)}\right).$$

A numerical example: probability of a trajectory

Consider the trajectory:

$$\mathcal{X}^{(0)} = (\mathit{sunny}, \mathit{low}), \qquad \mathcal{X}^{(1)} = (\mathit{rain}, \mathit{high}), \qquad \mathcal{X}^{(2)} = (\mathit{rain}, \mathit{high}).$$

Suppose that the DBN specifies:

$$P(\mathcal{X}^{(0)} = (\mathit{sunny}, \mathit{low})) = 0.30,$$
$$P(\mathcal{X}^{(1)} = (\mathit{rain}, \mathit{high}) \mid \mathcal{X}^{(0)} = (\mathit{sunny}, \mathit{low})) = 0.40,$$
$$P(\mathcal{X}^{(2)} = (\mathit{rain}, \mathit{high}) \mid \mathcal{X}^{(1)} = (\mathit{rain}, \mathit{high})) = 0.50.$$

Probability of this complete trajectory

$$P\big((\mathit{sunny}, \mathit{low}), (\mathit{rain}, \mathit{high}), (\mathit{rain}, \mathit{high})\big)$$
$$= 0.30 \cdot 0.40 \cdot 0.50 = 0.06.$$

The model assigns probability 0.06 to this specific evolution of the system over the three time steps.

Learning conditional distributions

Learning conditional distributions

What a Bayesian network specifies

Structure

Which variables are directly related:

$$Parents(X_i).$$

Usually designed from domain knowledge or learned from data.

Local probability models

The conditional distribution for each node:

$$P(X_i \mid Parents(X_i)).$$

Specified through probabilities or other model parameters.

Question for this section

How do we obtain the numerical probabilities or parameters of the local conditional models?

Where do the probabilities come from? I

Parameter learning / density estimation

Given a network structure, parameter learning estimates the conditional distributions

$$P(X_i \mid Parents(X_i))$$

from available data. More generally, this is a form of density estimation.

Possible sources of the parameters

- **Domain experts:** probabilities are elicited from expert knowledge;
- **Mechanistic models:** parameters follow from physical or domain-specific equations;
- **Data:** parameters are estimated from observations.

Where do the probabilities come from? II

Common learning approaches

- **Maximum likelihood:** choose the parameters that make the observed data most likely;
- **Bayesian learning:** combine a prior belief about the parameters with the observed data;
- **EM:** estimate parameters when some variables are hidden or some values are missing.

See Russel & Norvig, 4th Ed, Chapter 21 *Learning Probabilistic Models*

Beyond Bayesian networks

Beyond Bayesian networks

Probabilistic graphical models (PGMs)

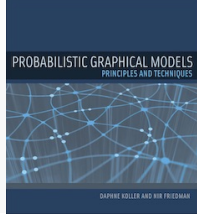
Use graphs to represent a joint probability distribution compactly.

Bayesian network	**Markov network**
directed graph	undirected graph
conditional distributions	compatibility factors
causal/generative direction	symmetric relations or constraints

When is an undirected model natural?

When no direction of influence is intended, for example:

- neighbouring pixels tend to have similar colours;
- two people tend to hold compatible opinions.



Markov networks

Markov network

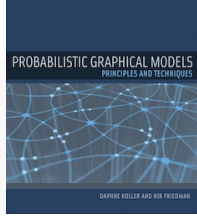
Represents a joint distribution through local compatibility relations, rather than through conditional probability tables.

Markov network

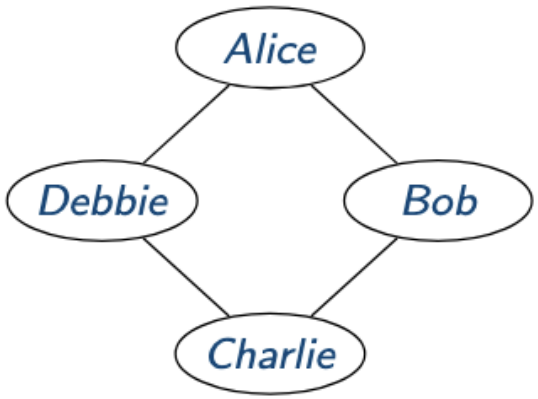
A Markov network is an undirected probabilistic graphical model. Its joint distribution is defined from nonnegative factors, or compatibility functions:

$$P(\mathbf{X}) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \phi_C(\mathbf{X}_C),$$

where Z is a normalization constant.



Markov networks: example I



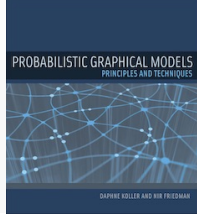
Misconception network

$\phi_1(A,B)$			$\phi_2(B,C)$			$\phi_3(C,D)$			$\phi_4(D,A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	b^1	c^1	100	c^1	d^1	1	d^1	a^1	100

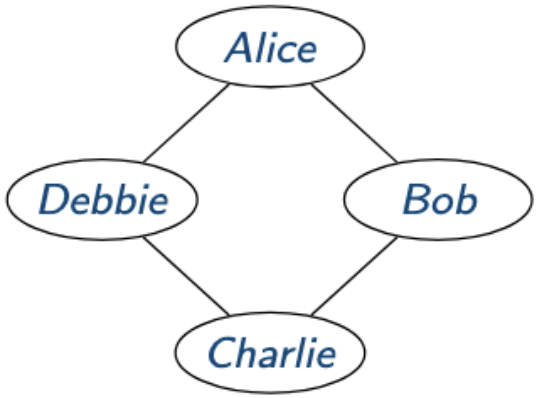
compatibility factors denoting affinity

How to read the example

- Node: a person whose variable can take two possible values (e.g., two possible opinions).
- Edge: compatibility factor; larger factor more compatible (these values are **scores**, not probabilities).



Markov networks: example II



Misconception network

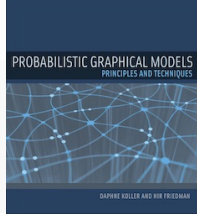
$\phi_1(A,B)$			$\phi_2(B,C)$			$\phi_3(C,D)$			$\phi_4(D,A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	b^1	c^1	100	c^1	d^1	1	d^1	a^1	100

compatibility factors denoting affinity

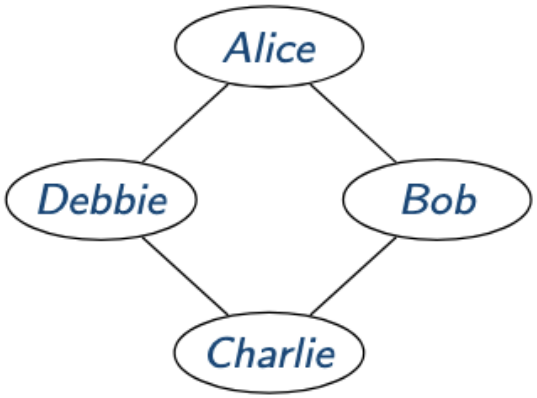
Suppose that a^0 and b^0 mean "in favour", whereas a^1 and b^1 mean "against".

The pair ($Alice = a^0$, $Bob = b^0$) receives the highest compatibility score.

Thus, this model considers it relatively plausible that Alice and Bob are **both in favour**.



Markov networks: example III



Misconception network

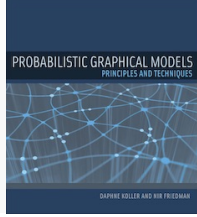
$\phi_1(A,B)$			$\phi_2(B,C)$			$\phi_3(C,D)$			$\phi_4(D,A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	b^1	c^1	100	c^1	d^1	1	d^1	a^1	100

compatibility factors denoting affinity

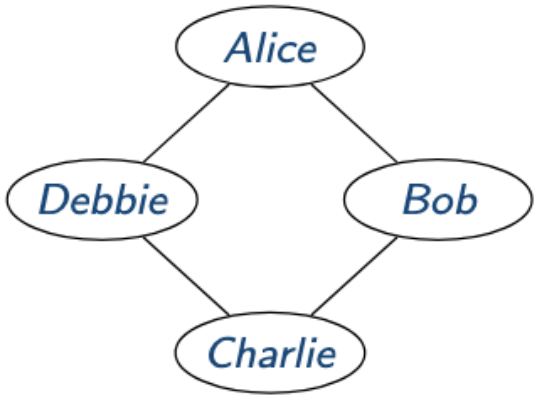
From local scores to a global score

For a complete assignment (a, b, c, d), multiply the compatibility factors:

$$w(a,b,c,d) = \phi_1(a,b)\,\phi_2(b,c)\,\phi_3(c,d)\,\phi_4(d,a).$$



Markov networks: example IV



Misconception network

$\phi_1(A,B)$			$\phi_2(B,C)$			$\phi_3(C,D)$			$\phi_4(D,A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	b^1	c^1	100	c^1	d^1	1	d^1	a^1	100

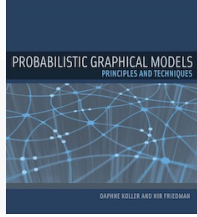
compatibility factors denoting affinity

A globally compatible assignment

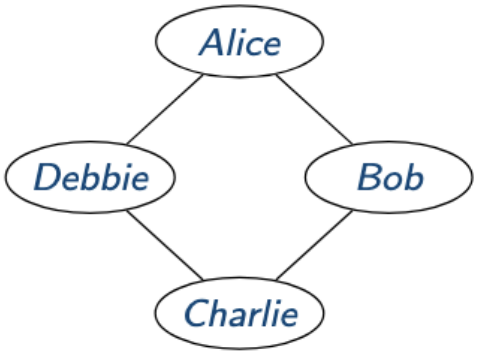
$$(a^0, b^1, c^1, d^0)$$

$$5 \cdot 100 \cdot 100 \cdot 100 = 5{,}000{,}000$$

It satisfies three strong local preferences: $b^1 = c^1$, $c^1 \neq d^0$, and $d^0 = a^0$.



Markov networks: example V



Misconception network

$\phi_1(A,B)$			$\phi_2(B,C)$			$\phi_3(C,D)$			$\phi_4(D,A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	b^1	c^1	100	c^1	d^1	1	d^1	a^1	100

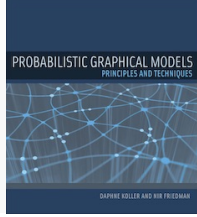
compatibility factors denoting affinity

A less compatible assignment

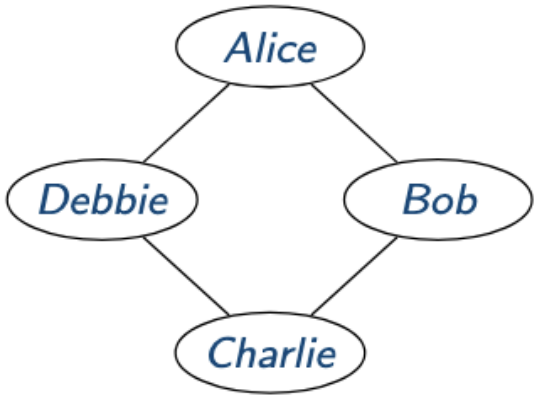
$$(a^0, b^0, c^0, d^0)$$

$$30 \cdot 100 \cdot 1 \cdot 100 = 300{,}000$$

Alice and Bob match, but Charlie and Debbie have the same value, which ϕ_3 considers incompatible.



Markov networks: example VI



Misconception network

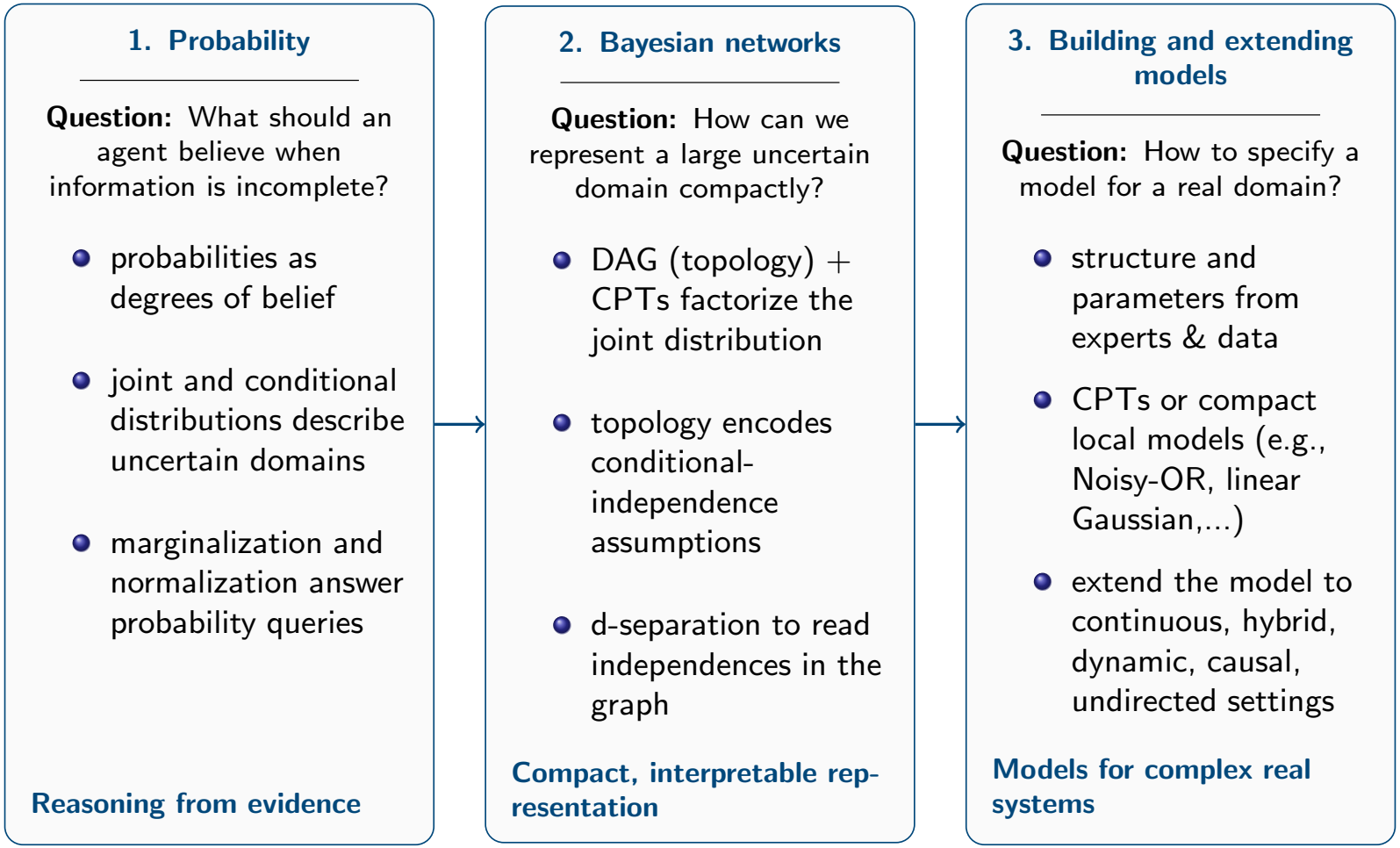
$\phi_1(A,B)$			$\phi_2(B,C)$			$\phi_3(C,D)$			$\phi_4(D,A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	b^1	c^1	100	c^1	d^1	1	d^1	a^1	100

compatibility factors denoting affinity

A Markov network combines local compatibility relations into a global score. After normalization, these scores become probabilities:

$$P(a,b,c,d) = \frac{w(a,b,c,d)}{Z}.$$

Summary so far



Next: how can we answer probability queries efficiently from these representations?

Suggested exercises from Russel & Norvig, 3rd Ed.

- 14.8 (car diagnosis), (a)-(d)
- 14.11 (nuclear plant)
- 14.12 (telescope problem)
- 14.14 (wrongful conviction)

Questions?

